Total no. of Pages: 02 — Roll no.........
Mid Semester Exam — Mar-2020
Fourth Semester — B.Tech.COE
CO202 Database Management System
Duration: 1.5 Hrs. — Max Marks: 30

NOTE: Attempt all the questions. Assume the missing data if any.

Total no. of Pages: 02 — Roll no.........
End Term Exam — May-2019
Fourth Semester — B.Tech. COE
CO 202 Database Management System
Duration: 3 Hrs. — Max Marks: 40

NOTE: Attempt all the questions. Assume the missing data, if any.

-104-

Total no. of Pages: 02 — Roll no.........
Mid Semester Exam — Mar-2019
Fourth Semester — B.Tech.COE
CO202 Database Management System
Duration: 1.5 Hrs. — Max Marks: 30

NOTE: Attempt all the questions. Assume the missing data if any.

Total No. of Pages 2 — Roll No. ....
FOURTH SEMESTER — *B.TECH (CSE)*
END SEMESTER EXAMINATION — *(MAY-2018)*

CO 202 DATABASE MANAGEMENT SYSTEM
Time: 3 Hours — Maximum Marks : 40

Note: Attempt any 5 questions. Question no. 1 is compulsory.
Assume suitable missing data, if any.

Total No. of Pages 2 — Roll No. ....
FOURTH SEMESTER — *B.TECH (CSE)*
MID SEMESTER EXAMINATION — *(MARCH-2018)*

CO 202 DATABASE MANAGEMENT SYSTEM
Time: 1.5 Hours — Maximum Marks : 30

Note: Attempt all questions. Assume suitable missing data, if any.

Total No. of Pages 2 — Roll No. ....
FOURTH SEMESTER — *B.TECH (CSE)*
END SEMESTER EXAMINATION — *(MAY-2018)*

CO/SE 211 DATABASE MANAGEMENT SYSTEM
Time: 3 Hours — Maximum Marks : 70

Note: Attempt any 5 questions. Question no. 1 is compulsory.
Assume suitable missing data, if any.

Total no. of pages 2 — [illegible]
FORTH SEM — [illegible]

MID SEMESTER EXAMINATION — [illegible]

CO/IT-202 DATABASE MANAGEMENT SYSTEM

Time: 1 Hour 30 min — Max. Marks : [illegible]
Note : Answer any four
Assume suitable missing data, if any.

Total no. of pages 2 — Roll No:....................
FORTH SEM — B.TECH(CO/SE)

MID SEMESTER EXAMINATION — MARCH 2016

CO/SE-211 DATABASE MANAGEMENT SYSTEM

Time: 1:30 Hours — Max. Marks : 20
Note : Answer all Questions
Assume suitable missing data, if any.

Total No. of Pages 2 — Roll No. ....
FOURTH SEMESTER — *B.TECH (CSE)*
END SEMESTER EXAMINATION — *(MAY-2019)*

CO/SE-211 DATABASE MANAGEMENT SYSTEM
Time: 3 Hours — Maximum Marks : 70

Note: Attempt any 5 questions. Question no. 1 is compulsory.
Assume suitable missing data, if any.

Total no. of pages 2 — Roll No: DTU/2K15/CO/049
Forth SEM — B.TECH(IT/CO)

END SEM EXAMINATION — May 2017

CO/IT-202 DATABASE MANAGEMENT SYSTEM

Time: 3 hr — Max. Marks : 70
Note : Answer any five
Assume suitable missing data, if any.

Total no. of Pages: 02 — *Roll no.........*
Supplementary Exam — Sept-2019
Fourth Semester — B.Tech. COE
CO 202 Database Management Systems
Duration: 3 Hrs. — Max Marks: 4

NOTE: Attempt any four questions. Assume the missing data, if any.

Total No. of Pages 2 — *Roll No. ......*
FOURTH SEMESTER — **B.Tech. [CO/SW]**

END SEMESTER EXAMINATION — ***MAY-2012***

CO/SW-211 DATABASE MANAGEMENT SYSTEM

Time: 3:00 Hours — *Max. Marks : 70*

Note : Question No. *ONE* is compulsory.
Answer any *FOUR* questions from remaining.
Assume suitable missing data, if any.

Date / / Page No.

UNIT-1: INTRODUCTION

Q1. (a) What is DBMS? What are the characteristics of DBMS? 2017M 3

A DBMS is a collection of programs that enables a user to maintain and access a collection of interrelated data called a database. It is a general-purpose software system that facilitates the process of defining, manipulating, constructing and sharing databases among various users and applications.

(c) What are the characteristics of DBMS? 2016 E 3

CHARACTERISTICS

→ Self describing nature of DBMS.
→ Insulation b/w programs and data, and data abstraction.
→ Support of multiple views of data.
→ Sharing of data and multiuser transaction processing (concurrency).
→ Ensures data integrity and consistency.
→ Security mechanism to control access of data.
→ Allows for data backup and recovery.

Q1. Answer the following questions in brief (2x5=10)
a) Compare the DBMS with file processing system. 2019 E

Q2. List four significant differences between a file processing system and a DBMS? 2020M (4)

Q6 Write short notes (Any two): 2018E
a. File Processing System vs. Integrated Database

	FILE PROCESSING SYSTEM	DBMS
1.	There is no builtin mechanism for creating backups of data and recovering data in event of failure	It includes features for creating multiple backups and restore points incase of problem.
2.	It may lead to data redundancy as the same info. may be duplicated in multiple files ⇒ waste storage	It avoids redundancy by storing data in a centralised locⁿ and relationship b/w tables to avoid dup
3	Here, multiple users cannot access the same file at the same time, leading to delays, and data conflicts	It allows concurrent access by locking, transaction mgmt etc allowing simultaneous access without conflicts.
4.	There is not a special mech for controlling access to data	It has features for controlling access to data such as acc., roles and permissions ensuring only auth. user have access to sensitive data.
5.	Inconsistent due to lack of integrity constraints	Ensures through use of constrains, triggers etc.

(b) What is DBMS? Explain the architecture of DBMS. 2016E 5

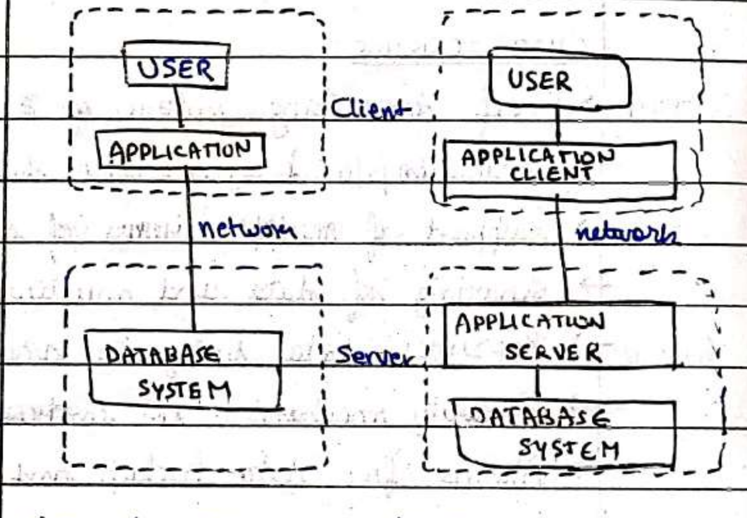
Q1 (a) Explain the concept of DBMS. Discuss its overall structure with diagram. 2018M

Q2. What is database architecture? Also explain data independence in detail. 2 2016M

(b) Describe the architecture and various component of database system? 8 2017E

Database architecture refers to the overall design and structure of a database system, including the logical and physical organization of data, hardware and software components used to implement the DB. It includes its structure and Database applications are usually partioned into two types:

- TWO-TIER: the application resides at the client machine where it invokes database system functionality at the server machine
- THREE-TIER: the client machine acts as a front end and doesnt contain DB calls
  → The client end communicates with an app server, usually through a form interface
  → The app. server in turn communicates with a DB system to access data.



a) Two-tier architecture b) Three-tier architecture

ii. Draw the 3 schema architecture of DBMS system and label it. 2018E

The goal of the 3 schema arch is to separate the user application from the physical database. It can be defined at the following 3 lvls.

1. Physical Schema: Internal schema describes the physical storage of database. This uses a physical data model and describes the complete details of data storage and access paths for the DB.

2. Conceptual Schema: Logical schema describes the structure of DB. It hides the details of physical storage structures and concentrates on describing entities, data-types, relationships, user operations and constraints. Its based on conceptual schema design in high lvl data model usually.

3. External Schema: View level includes a number of external schemas / user views. Each schema describes the part of DB a specific user is interested in and hides the rest of DB.

Date / /
Page No.

Attempt the following: (7X2=14)
Discuss the types of database. 2019E(a)

Types of DB acc to their usage and structure :

1.) HIERARCHICAL DB
These DBs organize data in a tree-like structure, where each record has one parent and many children. The relationships b/w records are one-to-many.
It is best suited for applications that involve a lot of data with a well-defined structure, such as a companies organizational chart.

2. NETWORK DB
They are similar to hierarchical but also allow records to have multiple parents. The relationship b/w records are many-to-many.
They are best suited for apps with complex data reln like CAD systems.

3. RELATIONAL DB
They organize data in tables, with each table representing a single entity or relationship. The relationship b/w tables are established using keys.
It is widely used and is best suited for appl's that require and involve large amounts of data and complex queries.

4. OBJECT-ORIENTED DB
It store data in objects, which consist of attributes and methods
It is best suited for applications that involve complex data structures, such as multimedia or scientific data.

Date / /
Page No.

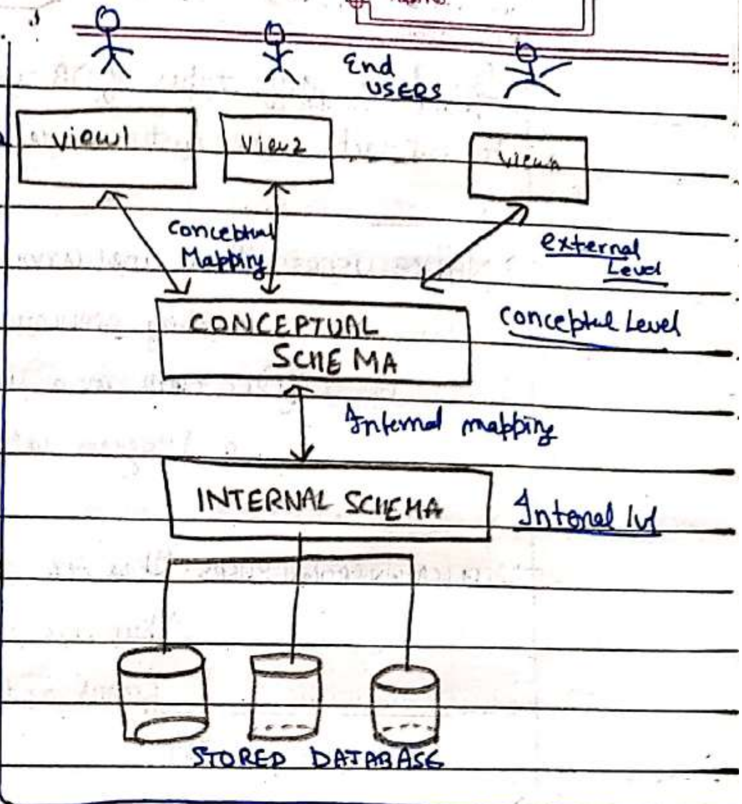
Q2. (a) Explain logical and physical indepenance. 2017M

DATA INDEPENDANCE

It refers to the capacity to change the schema at one level of a database system without having to change the schema at the next higher level.

1. LOGICAL DATA INDEPENDANCE refers to the capacity to change conceptual schema w/o having to change ext schema. It may be done to expand the DB, change constraints or remove DB.

2. PHYSICAL DATA INDEPENDANCE is the capacity to change the internal schema w/o having to change the conceptual schema (as well as external schema)



Q5 (a) Explain the concept of database schema with diagram. 2018M

A database schema is a blueprint of structure that defines how a DB is organised and the relationship between its components. It specifies the tables, fields, keys and other objects that make up the DB and how they are interlinked.

It serves as a framework for organizing and accessing data stored in it

c) Explain the different levels of data abstraction. Same

1.) Physical (Lowest): how data is stored in disk
2.) Logical: relation and rules in b/w data, data type, constraint. Phy is abstracted
3.) View level (Highest): how its shown to end user. Hides underlying complexity of logic, phy.

Q1. (a) What is the difference between security and Integrity? 2016E ?

SECURITY	INTEGRITY
It is concerned with DB protection from unauth-access and misuse and ensures only auth-users can access/modify the data	It is concerned with maintaining the accuracy and consistency of data and ensures that the data entered follows rules and constraints
It is acheived through authorization, authenti-cation, encryption and backup/recovery	It is achieved through data validation, integrity constraints and transaction management.

Date / Page No.

(b) What are different types of database interfaces. Classify them with example. (2X3=6)

2018 M

1. MENU BASED INTERFACES: These interfaces present the user with lists of options (called menus) that lead the users through the formulation of a request.
Eg.) pull-down menus in web-based interfaces.

2. FORM-BASED INTERFACES: A form-based interface displays a form to each user. Users can fill out all of the form entries to insert new data ~~or~~ etc.
Eg.) Oracle Forms.

3. GRAPHICAL USER INTERFACES (GUIs): A GUI typically displays a schema to the user in diagrammatic form. The user then can specify a query by manipulating the diagram. They, in many cases utilize both forms and menus.
Eg.) Microsoft Access

4. NATURAL LANGUAGE INTERFACES: These interfaces accept requests written in english or some other lang. and attempt to understand it.
Eg.) Siri / Alexa.

5. SPEECH I/O: Apps with limited vocab such as, inquiry for telephone dir, flight arrival/departure and bank use for ordinary folks to access.
Eg.) Voice-activated assistants like Echo.

6. INTERFACES FOR PARAMETRIC USERS: used for parametric users like tellers, who have a small set of instructions that they perform repeatedly.
Eg.) Func. keys in terminal can be programmed to initiate various commands.

7. INTERFACES FOR DBA: They contain privileged commands that can be used only by the DBA's staff. Ex) SQL*Plus for Oracle DB, which allows DBA to manage and manipulate database.

Date / /
Page No.

[b] Discuss the various database languages and interfaces. 2012E

~~SQL~~ → DML: discussed later
→ DDL: discussed later

→ Data Control Language (DCL)
It contains commands used to control the access of data stored in DB
Grant : grant permission to user for DB access.
Revoke : take back granted permission from user.

→ Transaction Control Language (TCL)
It contains commands used to ~~s~~manage the changes made by DML statements:
Commit : to permanently save the transaction.
Rollback: to undo the t/r under process.
Savepoint: to ~~t~~ temporary save a t/r so that user can rollback if needed.

→ Procedural language (PL)
PL is used to define procedural logic that can be executed within the DB. It allows users to create stored procedures and triggers
Eg) PL/SQL is an example of PL.

→ Object Relational Database Language (ORDBL). It is used to support object oriented structures such as classes and objects and perform OO operation like polymorphism and inheritance.
Eg) SQL3 is an example of ORDBL.

Date / /
Page No.

ii. Briefly explain the classification of database users. 2018E

→ NAÏVE USERS: These are unsophisticated users ~~which~~ who interact with the system by invoking APPLICATION PROGRAMS that have been written previously.
Eg:) a clerk in a university who needs to add new instructor uses a program called new_hire.

→ APPLICATION PROGRAMMERS: These are computer professionals who write application programs. They use tools like RAD that enable them to construct forms and reports with minimal programming effort.
Eg:)

→ SOPHISTICATED USERS: They interact with the system without ~~writing~~ writing programs but instead, they form their requests either using query language or tools like data analysis s/w.
Eg:) An analyst who submits queries to explore data in DB

→ SPECIALIZED USERS: They write specialized DB apps that do not fit in traditional data processing ~~sw~~ framework
Eg:)

e) Database Administrator 2020M

→ DATABASE ADMINISTRATOR:
A DBA is a specialised ~~user~~ type of user who is responsible for the management and maintenance of a database system.
They play a crucial role in ensuring the reliability, availability and security of the DB system, as well as ensuring that it meets the needs of the users and the organization as a whole

Date / / Page No.

(b) DBA 2016M

Q1 (a) Explain the role and responsibility a database administrator? 8 2017E

A DBA's role and responsibilities include administrating the resources since they have central control over the system. It includes:

→ Schema Defination: The DBA creates the original database schema by executing a set of DDL statements.

→ Storage ~~and~~ structure and access-method defination.

→ Schema and physical-organization modification: The DBA carries out changes to the schema and physical organization to reflect the changing needs of the organization, or to alter the physical organization to improve performance.

→ Granting of authorization for data access: By this, they can regulate which parts of the database various users can access. The auth. info is kept in a system structure that DBMS consults whenever someone tries to access.

→ Routine maintenance: this includes

- Periodically backing up the database, either onto tapes or onto tapes or onto remote servers, to prevent loss of data in case of disasters such as flooding.
- Ensuring that enough free disk space is available for normal operations, and upgrading disk space as required.
- Monitoring jobs running on the database and ensuring that performance is not degraded by very very expensive tasks submitted by some users.

1(a) What are the responsibilities of ~~DBA~~ and database designers? 2012E

They (DBD) are responsible for identifying the data to be stored in DB and for choosing appropriate structures for represent and store data. They communicate with all prospective database users in order to understand their requirements and to create a design that meets these requirements. They are often the staff of DBA and are assigned responsibility.

Date / /
Page No.

UNIT-3

7 Explain the following
(a) Functional Dependencies 2016M

A functional dependancy, denoted by $X \rightarrow Y$, b/w two sets of attributes X and Y that are subsets of R specifies a constraint on the possible tuples that can form a relation state r of R. The constraint is that, for any two tuples t_1 and t_2 in r that have $t_1[X] = t_2[X]$, they must also have $t_1[Y] = t_2[Y]$

Thus, it is a constraint b/w two sets of attributes from the database.

Q1 Attempt the following: (7X2=14)

i. List the advantages of database environment.

1.) DATA INTEGRATION: Centralized DB allows easier integration of data for storing and mgmt

2.) DATA SHARING: Multiple users can access and share data simultaneously.

3.) It allows for data consistency and reduce redundancy while improving data integrity.

4.) DATA SECURITY: has security mechs for controlling access, modification and distribution

5.) AVAILABILITY AND ACCESS: Data is always available along with tools for querying and manipulating data, improving productivity and dec making/decision

6.) IMPROVED PERFORMANCE: A DB environment can provide fast access to data, improving the performance of apps.

d) Mapping Cardinality 2020M

Mapping cardinality is used to express the number of entities to which another entity can be associated via a relationship set. It is most useful in describing binary relationship sets.

It can be of the of the following types:

→ one to one

→ one to many

→ many to one

→ many to many.

DATA MODELLING USING ER MODEL

Date / / Page No.

Q1. (a) What are terms associated with ER model. Draw extended ER diagram for a University employee system. 2016M 3

Terms associated with an ER model are:

→ ENTITY: It is the basic object that the ER model represents. It can be said as a "thing" or "object" in the real world that is distinguishable from all other objects, or it may be having a conceptual existance (Eg. job, university course).

→ RELATIONSHIP: A relationship is an association among several entities. It describes how entities are connected or related to one another.

∴ An entity set is a set of entities of the same type that share the same properties or attributes.

∴ A relationship set is a set of relationships of the same type so, if $E_1, E_2, \ldots E_n$ are entity sets, R is a subset of $\{(e_1, e_2 \ldots e_n) \mid e_1 \in E_1 \ldots e_n \in E_n\}$.

→ ATTRIBUTES: It is the properties or char. of an entity. An attribute for an entity set is a function that maps from the entity set into the domain.

2017M (ii) cardinality

⇉ CARDINALITY: It refers to the number of instances of one entity that can be associated with a single instance of another entity, such as one customer placing multiple orders.

...llowing: 2017E 8

(ii) Weak entity set and strong entity set

WEAK ENTITY SET: An entity set which does not have a primary key is reffered to as a weak entity set. It must relate to the identifying set via a total, one-to-many relh set from the identifying to weak entity set.

STRONG ENTITY SET: Strong entity set refers to a collection of entities that can be uniquely identified by their attributes. It can exist independently of the other entity sets, can be uniquely identified by its own attributes.

V

UNIVERSITY EMPLOYEE SYSTEM

2019E v. Total participation vs. Partial participation

Total Participation indicates that every entity in a relationship must be associated with atleast one entity in the other entity set. So, the participation of an entity in a relationship is mandatory. It is denoted by placing a double line between the entity and relationship in ER diagram.

Partial Participation refers than an entity in a reln may or may not be associated with an entity in another entity set. So the participation is partial. It is denoted by placing a single line b/w entity and reln in ER diag.

Ex.) In a "University DB", ~~en~~ 'Student' has total participation with 'enrollment'
"instructor" has partial " with " course offering "
(all inst dont teach particular subject)

Date / / Page No.

(b) Draw E-R diagram for a library system which is serving students and faculty of DCE. It is having book bank, reference section and reports collection. 2017M 3

iii. Composite vs. Atomic attributes 2018E

Atomic attributes are indivisible and cannot be divided further into meaningful components

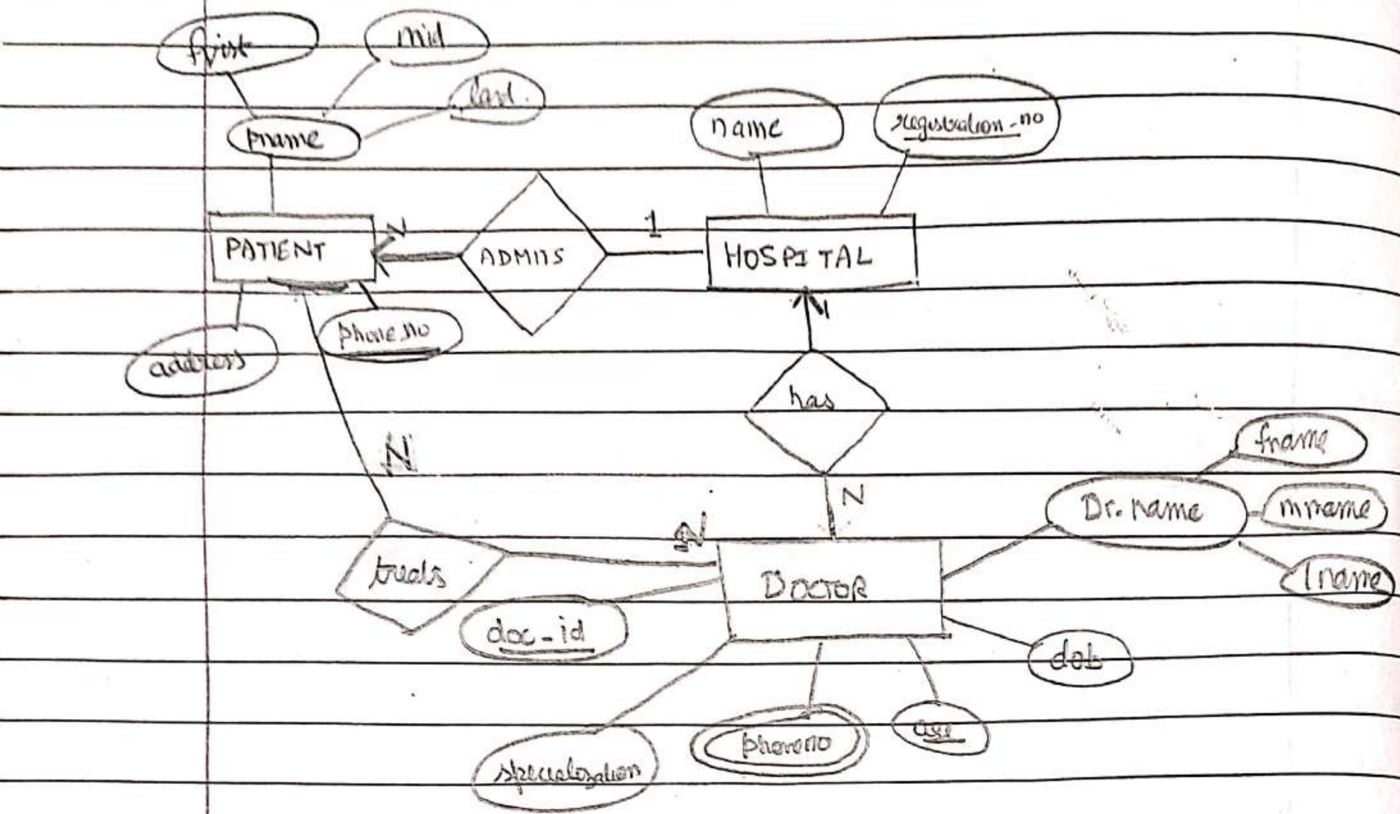
Eg:) age, height etc.

Composite attributes can be further subdivided into smaller sub-attributes or components..

Eg) A person's address can be subdivided into street-name, city, state and pin code

Date / /
Page No.

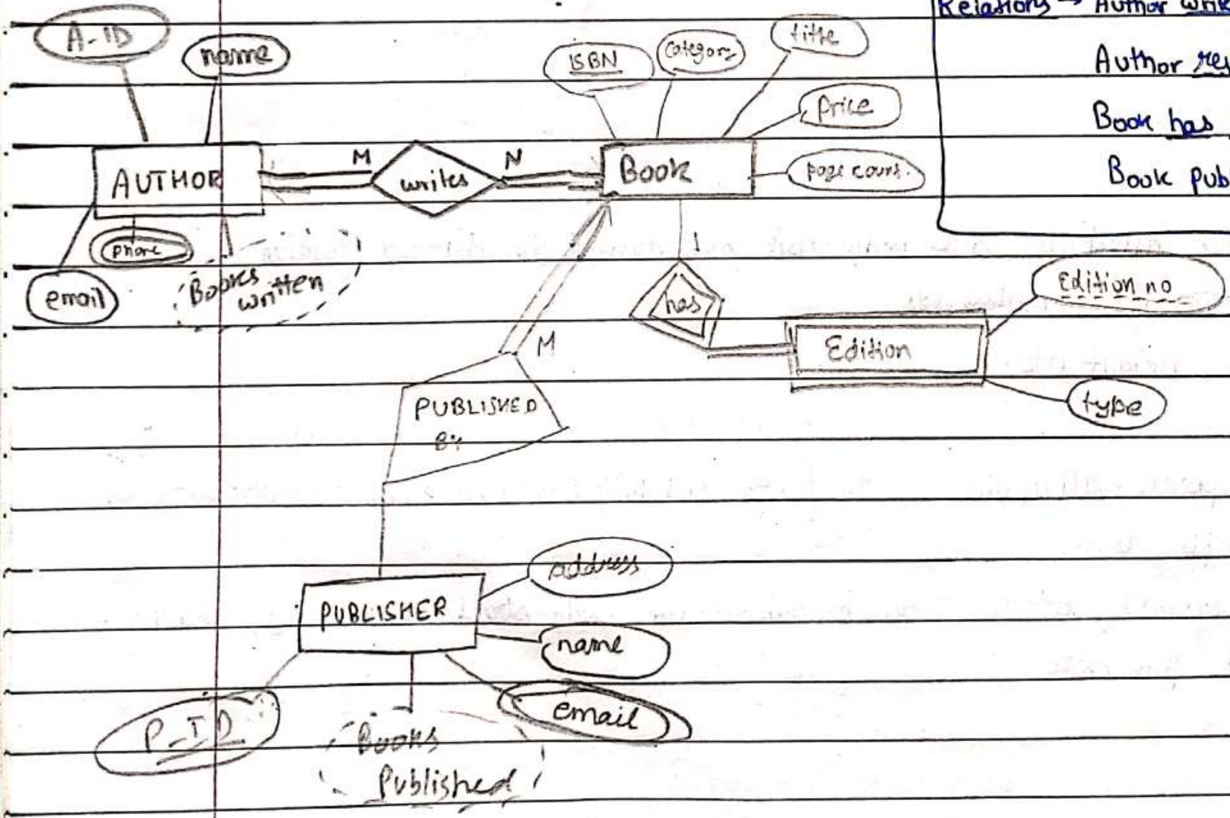
Q2. (d) Draw ER diagram for the hospital information system, 2017E 8



(c) Construct an ER diagram to model online book store. Map/con... relational model. 2016 E

Entities → AUTHOR, BOOK, PUBLISHER, (Strong)
EDITION (weak)

Relations → Author writes Book (M:N)
Author revises Book (M:N)
Book has Edition (1:M)
Book published by Publisher (M:1)

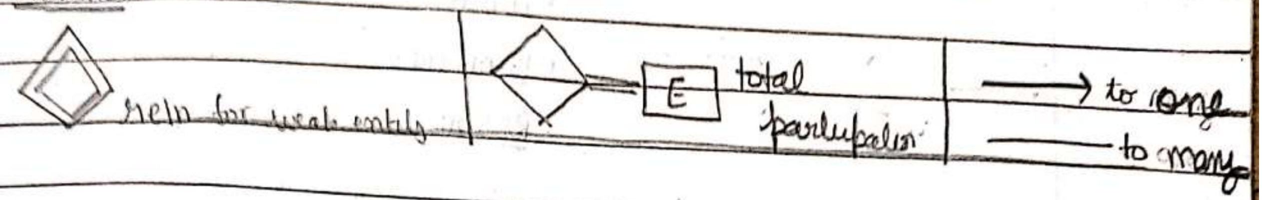


Q2 What is the need of using ER diagram. Explain its various components. Discuss the notation used to draw an ER diagram. Consider the following problem statement:

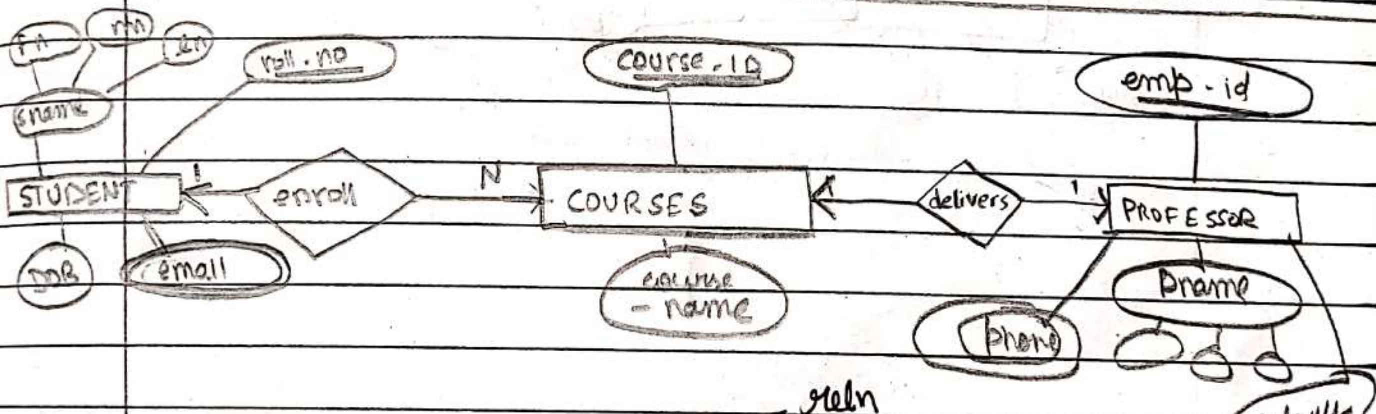
In a university, a Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course.

Identify the entities, attributes and relationships and draw an ER diagram. 2019E (14)

NOTATION



ER diagram



reln

ENTITIES:	STUDENT	enrolls	COURSES	delivers	PROFESSOR
ATT	Sname		CourseID		emp-id
	rollno		course name		pname
	email				Salary
	DOB				phone.

Q2 (a) A university registrar's office maintains data about the following entities:

1. courses, including number, title, credits, syllabus, and prerequisites;
2. course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
3. students, including student-id, name, and program;
4. instructors, including identification number, name, department, and title.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

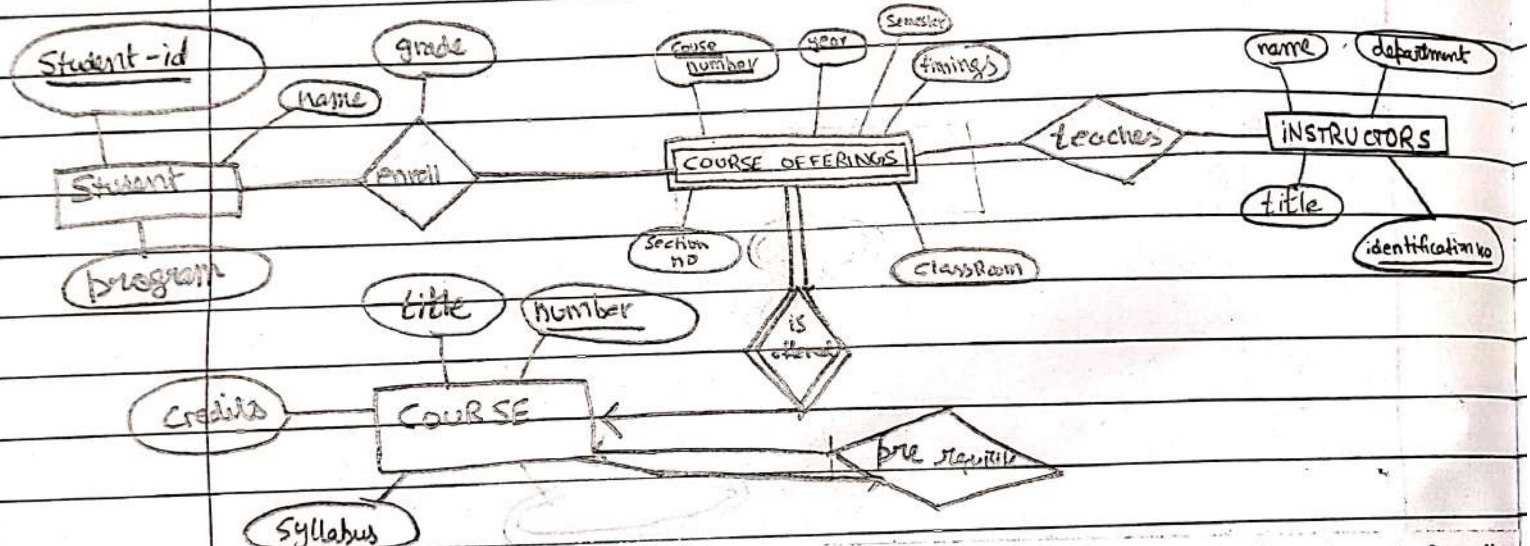
Identify the entities, attributes and relationships. Also, draw an ER diagram for the above problem statement. 2018E 14

Entities :

- Student : Student-id, name, program
- instructor : id-no, name, dept, title
- course : syll, c no, title, credits, prerequisites
- course off. : sec.no, year, sem, time, room

Relationship :

- ENROLL
- TEACHES
- IS OFFERED
- REQUIRES



Q2 (a) Consider the following set of requirements for college database:

A college contains many departments. Each department can offer any number of courses. Many instructors can work in a department. An instructor can work only in one department. For each department there is a Head. An instructor can be head of only one department. Each instructor can take any number of courses. A course can be taken by only one instructor. A student can enroll for any number of courses. Each course can have any number of students.

Identify the entities, attributes and relationships. Also, draw an ER diagram for the above problem statement. 2018E

Date / /
Page No.

2 A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game and the result of game. Design an ER schema diagram for this applications stating any assumptions you make, choose your favorite sport. 14

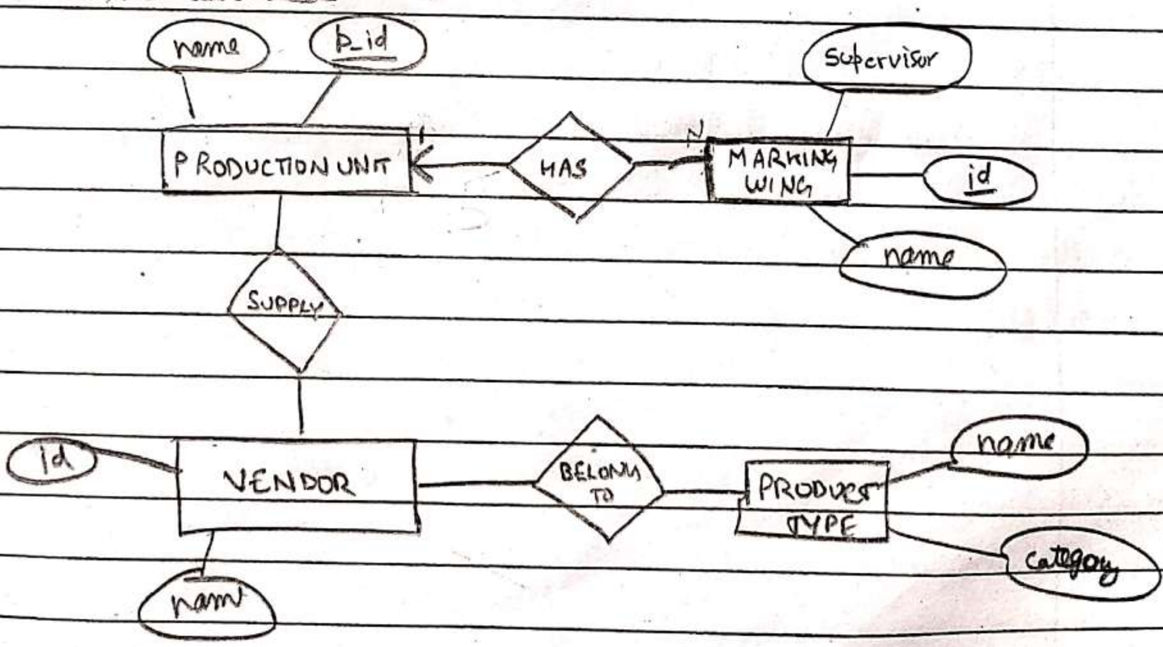
2012E

Q3. Draw an ER diagram for a garment manufacturing company. The entity includes warehouses, production units, marking wing, vendor and product types. Define the relationship between each of these entities and take the appropriate attributes. (6)

2019M

?

6



Date / /

Page No.

Q2 Consider the following set of requirements for a Company database that is used to keep track of employee's department and their project transcripts:
Company is organized into various Department with a unique name and a particular employee who manages the department. Department may have several locations. Start date for the manager is recorded. A department controls a number of Projects with a unique name, number and a single location. Company's Employee name, id, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by his/her dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's Dependent are tracked for health insurance purposes (dependent name, birthdate, relationship to employee).
Draw an ER Diagram for the above problem statement. 2018M (6)

Date / /
Page No.

(b) What are the steps to map E-R model to relational model? 2017M

The steps are:

1. Identify the entities and relationships in the ER diagram
2. Create a table for each entity, with columns for each attribute
3. Identify the primary key for each table, which will be used to link tables together.

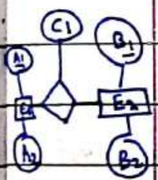
a.) Strong entities with simple attributes: They require only one table and its attributes will be the attributes of the entity set.

b.) " " composite " : They also require one table and simple attributes of the entity will be considered and not composite att.

c.) " " multivalued " : One table will contain all simple attributes and N tables will include PK and one of multivalued att.

d.) Binary relⁿ with CR → M:N : three tables will be reqd → two for each entity and one showing relⁿ b/w them.

4. Create a foreign key for one to many, many to one, one to one, linking.



e.) " " 1:N : two tables are reqd. | B1(PK) | B2 | C1 | A1(F) | | A1(P) | A2 |

f.) " N:1 : " " | A1(P) | A2 | C1 | B1(FK) | | B1(P) | B2 |

g.) 1:1 : two choices of 2 tables are there

h.) Binary relⁿ with weak entity set :

i.) Binary relⁿ - total participation constraint :

j.) N-Ary Relⁿ

5. Check for any business rules that may affect the design and modify the schema accordingly. Validate the relational model by checking for data integrity and consistency.

Q.3 (a) define the terms 2017M
(i) Entity set (ii) specialization
(v) Entity Integrity

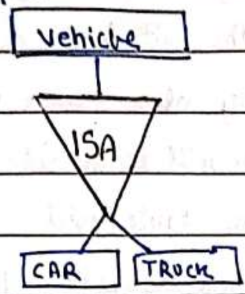
UNIT 2: RELATIONAL DATA MODEL AND LANGUAGE

Date / / Page No.

i) refer prev

ii) Specialization is a way to represent a subtype of an entity. It is a top down design process where we designate subgroupings within an entity set that are distinct from other entities in the set.
These subgroupings become lower level entity-sets and are having att. or participate in relh that do not apply to higher ~~set~~-lvl entity set.

Eg.)



v.) It is a database concept that ensures that each row in a table has a unique identifier. This constraint ensures that there are no duplicate rows in a table and that each row can be identified and accessed independently.

b) Derived attributes 2020M

Derived attributes are characteristics that can be calculated or inferred from other attributes in a database or data model. They are not stored but rather computed based on existing data.

Eg.) "Age" attribute can be calculated using "DOB" (and current date)

Q3. a) Explain generalization and serialization with suitable example. 2019S (2)
2020M (2)

b) Explain generalization and specialization with the help of suitable example. 2019S (3)

Generalization is a bottom-up process to combine a number of entity sets that share the same features. It is the inverse of specialization.

SERIALIZATION ??

Q3 (a) Explain relational algebra. What are the basic relational algebra operations. Discuss with example. Consider the following relational schema and answer the following queries using relational algebra: 2019E (8+6=14)

BOOKS(DocId, Title, Publisher, Year)
STUDENTS(StId, StName, Major, Age)
AUTHORS(AName, Address)
borrows(DocId, StId, Date)
has-written(DocId, AName)
describes(DocId, Keyword)

i. List the name of students who are older than 30 and who are not studying CS.
ii. List all books published by McGraw-Hill before 1990.

(c) Relational Algebra 2016M

7. Write Short Notes (any four)
Relational Algebra 2016E.

The relational algebra is a procedural query language. It consists of a set of operations which take relation(s) as I/P and produce new relations as output. It consists of functional operations like select, project union, rename, cartesian product, set diff. and joins etc.

i.) $\Pi_{StName}(\sigma_{Age>30}(STUDENTS)) - \Pi_{StName}(\sigma_{Major='CS'}(STUDENTS))$

ii.) $\sigma_{Publisher='McGraw-Hill' \wedge Year<1990}(BOOKS)$

OPERATIONS

DIVISION OPERATOR ($\div$)

Let r and s be relations on schemes R and S respectively where

$R = (A_1, A_2 --- A_m, B_1, B_2, --- B_n)$

$S = (B_1, B_2 ---- B_n)$

The result of $r \div s$ is a relation on schema

$R - S = (A_1, A_2 --- A_m)$

$$r \div s = \{ t \mid t \in \Pi_{R-S}(r) \wedge \forall u \in s\ (tu \in r) \}$$

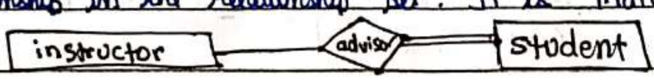
where tu means concat of t and u in single tuple.

Division operator can be applied only if "Attributes of B is proper subset of Attributes of A".

c) Explain the two types of participation constraint.

This constraint specifies the number of instances of an entity that are participating in the relationship type

→ TOTAL PARTICIPATION: every entity in the entity set participates in atleast one relationship in the relationship set. It is indicated by double line



Student participation is total
Instructor " is partial.

→ PARTIAL PARTICIPATION:
some entities may not participate in any relationship in the rel^n set

Q2. a) Consider the following relational schema
employee(empno, name, office, age)
books(isbn, title, authors, publisher)
loan(empno, isbn, date)

2020M

Write the following queries in relational algebra.

i) Find the names of employees who have borrowed a book published by TMH.

ii) Find the names of employees who have borrowed more than five different books published by TMH.

2019E

(5) Q1. Given the relational schemas:
employee(empno, name, office, age)
books(isbn, title, authors, publisher)
loan(empno, isbn, date)

Write the following queries in relational algebra.

a) Find the names of employees who have borrowed a book published by TMH.

b) Find the names of employees who have borrowed more th[an] five different books published by TMH.

i.) Π_{name} (employee $\bowtie$ (Π_{empno} (Loan $\bowtie$ (Π_{isbn} ($\sigma_{bublisher='TMH'}$ (books))))))

? ii.) Π_{name} (employee $\bowtie$ (Π_{empno} (Loan $\bowtie$ (Π_{isbn} ($\sigma_{publisher='TMH' \text{ AND } COUNT(isbn)>5}$ (books)))

Q3 (i) Consider the following relational schema:
Author (AuthorId, FirstName, LastName)
AuthorPub (AuthorId, PubId, AuthorPosition)
Book (BookId, BookTitle, Month, Year, Editor)
Pub (PubId, Title, BookId)

AuthorId in AuthorPub is a foreign key referencing Author. PubId in AuthorPub is a foreign key referencing Pub. BookId in Pub is a foreign key referencing Book. Editor in Book is a foreign key referencing Author (AuthorId).

(a) Determine the names of all authors who are book editors.
(b) Obtain the names of all authors who are not book editors.
(c) Get the names of all authors who have at least one publication in the database.

2018M

Write the above queries in Relational Algebra. (3X1=3)

a.) $\Pi_{firstName, Last Name}$ (Author $\bowtie_{Author ID = editor}$ Book)

b.) $\Pi_{first Name, LastName}$ (($\Pi_{AuthorID}$ (Author) $-$ Π_{Editor} (Book)) $\bowtie$ Book)

c.) $\Pi_{FirstName, LastName}$ (Author $\bowtie$ AuthorPub)

(b) Consider the following relational schema and answer the following queries using relational algebra:
Dealer (Dealer_no, Dealer_name, Address)
Part (Part_no, Part_name, Color)
Assigned_To (Dealer_no, Part_no, cost)

2018E

i. Find name of dealers that supply both red and yellow parts.
ii. Find names of dealers who supply all parts.
iii. Find names of dealers that supply whole red parts. (2X3=6)

i.) Π_{Dealer_name} ($\sigma_{color='Red'}$ (Dealer $\bowtie$ Assigned_To $\bowtie$ Part

ii.) $R_1 = \Pi_{Part\_Name}$ (Part)
$R_2 = \Pi_{Dealer\_name, Part Name}$ (Dealer $\bowtie$ Assigned_To $\bowtie$ Part)
$R_1 \div R_2$

iii.) $R_1 = \Pi_{Part\_name}$ ($\sigma_{color='Red'}$ (Part))
$R_2 = \Pi_{Dealer\_name, Part\_name}$ (Dealer $\bowtie$ Assigned_To $\bowtie$ Part)
$R_1 \div R_2$

Date / / Page No.

3[a] Consider the following relations for a database that keeps track of business trips of sales persons in a sales office:

SALES PERSON (SSN, Name, Start-Year, Dept-No)
TRIP (SSN, From_city, To_City, Departure, Date, Return-Date, Trip-ID)
EXPENSE (Trip-ID, Account #, Amount)

Specify the following queries in relational algebra: 2012E

(i) Give the details (all attributes of TRIP relation) for trips that exceeded $ 2000 in expenses.

(ii) Print the SSN of sales man who took trips to 'America'.

(iii) Print the total trip expenses incurred by the salesman with SSN = '237-66-1234'.

i) $\Pi(\text{TRIP} \bowtie (\sigma_{amount > 2000}(\text{EXPENCE}))$

ii.) $\Pi_{SSN}(\sigma_{To\text{-}city = 'America'}(\text{TRIP}))$

iii.) $\Pi_{amount}(\sigma_{SSN = '237-66-1234'}(\text{TRIP} \bowtie \text{EXPENSE}))$

Q1.Given the relational schemas: (2.5x2)

employee (person_name, street, city)
works (person_name, company_name, salary)
company (company_name, city)
manages (person_name, manager_name)

2019M

Give the expressions in relational algebra with explanation for the following queries:

(a) Find the names of all employees who live in the same city and on the same street as do their managers.
(b) Give all managers in this database a 20 percent salary raise.

a.) $\Pi_{person\_name}(\text{employee} \bowtie \text{managers} \bowtie$ $(\text{manager\_name} = \text{employee2.person\_name AND employee.street} = \text{employee2.street}$ $\text{AND employee.city} = \text{employee2.city})\ (\rho_{employee2}(\text{employee}))$

b.) $\text{managers} \leftarrow \Pi_{person\_name}(\sigma_{person\_name = manager\_name}(\text{manages}))$

$\text{Updated\_Salaries} \leftarrow \sigma_{works.person\_name = manages.person\_name}(\text{works} \bowtie \text{manages})$

$\text{u.Salary} \leftarrow \text{udated\_salaries.salary} * 1.2$

Q1. Given the relational schemas: 2019S (2.5x4)

ENROLL(s#, c#, section) – s# represents student number
TEACH (prof, c#, section) – c# represents course number
ADVISE (prof, s#) – prof is thesis advisor of s#
PRE_REQ(c#, pre_c#) – pre_c# is prerequisite course
GRADES (s#, c#, grade, year)
STUDENT(s#, sname) – sname is student name

Give queries expressed in relational algebra for the following queries:

a) List all students taking courses with Smith or Jones.
b) List all students taking at least one course that their advisor teaches.
c) List those professors who teach more than one section of the same course.
d) List the courses that student 'John' can enroll, i.e. has passed the necessary pre_requisite courses but not the course itself.

a.) $\Pi_{s\#}(\sigma_{sname='Smith' \vee sname='Jones'}(\text{STUDENT} \bowtie \text{ENROLL}))$

b.) $\Pi_{ADVICE\text{-}S\#}(\text{TEACH} \bowtie_{TEACH.prof = ADVICE.prof} \text{ADVICE})$

c.) $\Pi_{prof}((\sigma_{Count(x) > 1}(\text{TEACH}) \bowtie \Pi_{c\#, section}(\text{TEACH}) \bowtie \Pi_{s\#}(\text{ADVICE}))$

d.) $\Pi_{COURSES.C\#}(\sigma_{Student.sname = 'John'}$ $\wedge (\text{NOT EXIST}(\sigma_{enroll.s\# = student.s\#}$ $\text{AND enroll.s\#} = \text{PREREQ.pre\_c\#})$ $)$ $)$

?

Date / / Page No.

(b) Explain the operations of Relational Algebra. 2017M

SELECT OPERATION (σ)

The select operation selects tuples that satisfy a given predicate. The predicate appears as a subscript to σ and the relation arg. in the paranthesis after σ.

S ∴ $\sigma_{condition}$ (Relation - name)

Eg.)

Roll	Name	CGPA
3K31/01	Rachel	8.7
3K31/02	Narayan	9.5
3K31/03	Aoi	10

$\Rightarrow \sigma_{CGPA>9}(S) \Rightarrow$

Roll	Name	CGPA
3K31/02	Narayan	9.5
3K31/03	Aoi	10

PROJECT OPERATION (π)

The project operation is a unary operation which returns its argument rel^n, with certain attributes left out. We list the attributes those we wish to appear in the final result in the subscript to π and rel'n in paranthesis

Eg.) ∴ $\pi_{Roll, CGPA}(S)$

$\rightarrow$

Roll	CGPA
3K31/01	8.7
3K31/02	9.5
3K31/03	10

SET OPERATIONS

1. UNION OPERATION ($\vee$)

The UNION operation produces the tuples which are in either relation or both. It is denoted by R U S or R V S. Duplicate tuples are eleminated.

2. INTERSECTION OPERATION ($\wedge$)

The result of this operation, denoted by $R \cap S$/$R \wedge S$ is a relation which includes all tuples that are in both R and S

3. SET DIFFERENCE (–)

The result of this operation denoted by R-S, is a relation that includes all tuples that are in R but not in S

4. CROSS PRODUCT (x)

It produces a relation that has attributes of R_1 and R_2 and includes as tuples all possible combinations of tuples from R_1 to R_2.

Date / /
Page No.

RENAME OPERATION (ρ)

The rename operation can rename either the relation name or the attribute name or both as a unary operator. The general operation when applied to R of degree n can be denoted as:

$\rho_{S(B_1, B_2, ---B_n)}(R)$ or $\rho_S(R)$ or $\rho_{(B_1, B_2 -- B_n)}(R)$.

JOIN OPERATIONS. ($\bowtie$)

The Join Operation is used to combine related tuples two relations into single "longer" tuple.

$\therefore \sigma_{condition}(R \times S) = R \bowtie_{condition} S$

It can be:

→ Inner Join	→ Outer Join
→ Theta Join ($\bowtie_\theta$)	→ Left Outer Join (⟕)
→ Equi Join ($\bowtie_=$)	→ Right outer Join (⟖)
→ Natural Join ($\bowtie_{A.P = B.P}$)	→ Full outer Join (⟗)

AGGREGATE FUNCTION AND OPERATION

Aggregation operation in reln algebra

$$G_1, G_2 --- G_n \, \mathcal{G}_{F_1(A_1), F_2(A_2) --- F_n(A_n)}(E)$$

E is any relation algebra expression

→ $G_1, G_2 --- G_n$ is a list of attributes on which to group (can be empty)

→ Each F_i is an aggregate function

→ Each A_i is an attribute name

Date / / Page No.

(b) What are key, primary key, candidate key and super key? What is the difference between them, explain with example. 2 2016M

A key is an attribute or a set of attributes in a table that can help to uniquely identify a record in the database table.

Primary Key: It is a type of key that uniquely identifies each record in a table. There can only be one PK in a table and it cannot contain NULL values.

Candidate key: It is an attribute or a set of attributes that can qualify as a PK. There can be multiple CKs in a table. It is the minimal superkey.

Super key: It is a set of attributes in a table that can uniquely identify a record. It can be any combination of attributes including the PK.

Eg.) Student (Student_ID, Name, Age, Address)
∴ PK – Student_ID
CK – {Student_ID, Name}, {Name, Address} – –
SK – {Student_ID, Name, Age}, {Student_ID, Address} – – – –

b) Differentiate between primary key and partial key using example. 2020M (2)

PRIMARY KEY	PARTIAL KEY (Secondary Key)
It can uniquely identify each record in a table and each table has only one primary key	It does not uniquely identify each record in a table and can have multiple partial keys.
It cannot have have null values and is unique	It can have NULL values and is non unique.

Eg.) Student (ID, Name, Age, Add)
∴ Partial Key – Name, Age
Primary Key – ID.

iii. Define primary key, alternate key and candidate key with example. 2019E

Alt Key: It is a key which can be used to identify a record but is not a primary key. All candidate keys except PK is an alt key

Eg.) {ID, Name}, {Name, Add}

Date / /
Page No.

v. Discuss anomalies and its classification. 2018E

An anomaly refers to an unexpected or unintentional outcome that arises when data is manipulated or processed. They occur when data is inserted, deleted or updated leading to data inconsistency, loss or corruption.

1. INSERTION ANOMALY: It occurs when its not possible to add data to a table without also adding additional data that is not relevant to record

   Eg.) we must include nulls or other values if dept is not reqd in a new employee record.

   It can also occur in a scenario when inserting a new dept. having no employees.

2. DELETION ANOMALY: It occurs when deletion from a table also results in the unintended loss of other data that may be imp.

   Eg) If we delete from emp-dept a tuple then if he was the last employee of that dept, then info of the dept too is lost.

3. UPDATE/MODIFICATION ANOMALY: It occurs when updating data in a table results in inconsistencies or errors.

   Eg.) If manager of a dept change we have to update in records of all employees having that dept.

Date / / Page No.

Q5. (a) Explain the following Integrity rules of relational model : 8
(i) Entity integrity
(ii) Referential integrity
2017E

i:) Entity Integrity :
It states that the primary key of a table must be unique and that a table cannot contain a null value in P-K field but in other.

Eg.)

ID	Name
001	Ram
002	Shyam
null	Aoi

violates constraints

→

ID	Name
001	Ram
002	Shyam
003	Aoi

Correct table

iv. Explain referential integrity constraint with example. 2018E

ii:) Referential Integrity :
c) Referential Integrity 2020M

It is pertinent to FK and is enforced when FK references PK of another table. It states that

→ If a FK in table 2 refers to the PK of table 1 then either every value of the FK in table 2 must be available in PK value of table 1 or it must be NULL

→ A record in table 1 can not be deleted if if it doesnt exist in table 2.

→ A new value of in FK in table 2 cannot be added if it doesn't exist in table 1. Instead, NULL value can be added.

Eg.)

ID	Name	SUB-ID	
001	Aoi	CS01	✓
002	Rohit	CS07	X

STUDENT

→

SUB-ID	SUB-Name
CS01	OS
CS02	DBMS
CS03	ADA

SUBJECT

(e) Schema based constraints 2016M

Schema based constraints refers to set of rules that defines the structure and content of data in a database. They help maintain data integrity :

iii:) DOMAIN INTEGRITY : It states that the domain or type of an attribute must be within its domain.

iv.) KEY CONSTRAINTS : It specifies that in any relation
→ PK must not be NULL.
→ All the values in PK attribute must be unique.

Q3 (a) Discuss relational calculus. Differentiate between tuple relational calculus and domain relational calculus with example. 8 2018E

(d) Relational Calculus 2016E

Q6 Write short notes (Any two): 2019E
a. Relational calculus and it types

Relational Calculus is a non-procedural language (declarative) which is used to describe the retrieval of information from a relational database. (It tells what to do but no description about how to do) It uses mathematical predicate logic to retrieve results. There are two main types of relational calculus TRC and DRC:

TUPLE RELATIONAL CALCULUS (TRC)	DOMAIN RELATIONAL CALCULUS (DRC)
It is a formal language that uses First order logic to describe the desired result set.	It is a NP query language which is equivalent in power to TRC
It is used for selecting the tuples in a relation that satisfy the given predicate	The records are filtered on the basis of domains (so it selects attributes rather than whole tuples.
EXPRESSION: $\{t \mid P(t)\}$ $t \rightarrow$ resulting tuple $P(t) \rightarrow$ predicate used to fetch t	EXPRESSION: $\{\langle x_1, x_2, \ldots x_n \rangle \mid P(x_1, x_2 \ldots x_n)\}$ $x_1, x_2, \ldots x_n \rightarrow$ domain variables $P \rightarrow$ condition wrt DRC

Eg.) to find names of all customers having a loan ~~and an account at~~ the bank greater than \$2000

TRC $\rightarrow \{t \mid t \in loan \wedge t[amount] > 2000\}$

DRC $\rightarrow \{\langle c \rangle \mid \exists\, l, b, a\, (\langle c, l \rangle \in borrower \wedge \langle l, b, a \rangle \in loan \wedge a > 2000)\}$

SCHEMA

Loan (loan_no., branch_name, amount)

borrower (customer_name, loan_no.)

Q3 Write the sql queries for the relation Books(title,author,book_id,year_publication) 1x5

IT2019

a) Books(all details) whose author name has letter 'A' at the third place.
b) Book (only title) with the second largest year of publication.
c) Books (all details) published between 1990-2000 but not in year 1990 and 2000
d) Create a table called mybooks including only year and title.
e) Set the value of year to 2013 in mybooks (created above) where title starts with 'P'.

e.) UPDATE mybook SET year = 2013 WHERE title like 'P%'

~~Q4 Explain the components of DBMS with the help of a diagram~~ 5

a.) SELECT * from books where author LIKE '__A%'

b.) SELECT title from books ORDER BY year_publication DESC LIMIT 1 OFFSET 1

c.) SELECT * from Book where yearpub BETWEEN(1991 AND 1999)

d.) CREATE TABLE mybook (
year INT,
title VARCHAR (255)
); INSERT INTO mybook (year, title) SELECT year_pub, title from Book

P.T.O

(ii) Consider the following relational schema:
Student (rollNo, name, degree, year, sex, deptNo, advisor)
Department (deptId, name, hod, phone)
Course (courseId, cname, credits, deptNo)
Enrollment (rollNo, courseId, sem, year, grade)
a. Determine the departments that do not have any girl students.
b. Obtain the names of courses enrolled by student named Thomas.
c. Get the names of students who have scored 'O' in all subjects they have enrolled. Assume that every student is enrolled in at least one course.

2018M

Write the above queries in Tuple Relational Calculus. (3X1=3)

a.) $\{ d.name \mid department(d) \wedge \exists s\,(student(s) \wedge s.sex = 'F' \wedge s.deptNo = d.deptId) \}$

b.) $\{ c.name \mid course(c) \wedge (\exists s)(\exists e)\,(Student(s) \wedge enrollment(e) \wedge s.name = \text{"Thomas"}, \wedge s.rollNo = e.rollNo, \wedge c.courseId = e.courseId \}$

c.) $\{ s.name \mid student(s) \wedge (\forall e)((enrollment(e) \wedge e.rollNo = s.RollNo) \wedge e.grade = 'S')) \}$

(b) Consider the following schema and write queries in domain relational calculus: 2018E
loan(loan number, branch name, amount)
borrower (customer, loan)

i. Find all loan numbers for loans with an amount greater than \$1200.
ii. Find the names of all customers who have a loan from the Perryridge branch and find the loan amount. (6+2=8)

i.) $\{ \langle c \rangle \mid \exists l, b, a\,(\langle c, l \rangle \in borrower \wedge \langle l, b, a \rangle \in loan \wedge a > 1200) \}$

ii.) $\{ \langle c, a \rangle \mid \exists l\,(\langle c, l \rangle \in borrower \wedge \langle l, \text{"Perryridge"}, a \rangle \in loan) \}$

OR

$\{ \langle c, a \rangle \mid \exists l\,(\langle c, l \rangle \in borrower \wedge \exists b\,(\langle l, b, a \rangle \in loan \wedge b = \text{"Perryridge"})) \}$

(b) Differentiate the f
(i) DDL and DML
2017E

c) Explain DDL and DML. 2016E

DDL (Data Defination Language)

DDL provides commands for defination of rel^n schemas, deleting relations, and modifying relation schemas. It allows specification of not only a set of relations, but also information about each of relation including: Schema for rel^n, type of values, integrity constraints, set of indices to be maintained, security and auth infor and physical storage structure of each rel^n on disk.

It contains CREATE, ALTER, DROP, TRUNCATE etc.

DML (Data Manipulation Language)

It is known as query language
Its of 2 types:
DECLARATIVE: only tell what's needed?
PROCEDURAL: what is needed? how to get?

DML provides commands for manipulating data within the database. DML statements are used to insert, delete, update and retrieve data from tables in DB. They allow users to interact with data in a flexible and powerful way, making it easier to manage and maintain large amounts of information.

It contains INSERT, UPDATE, DELETE, SELECT etc.

5. Differentiate between: 2019E (4x2=8)
a) Procedural and non-procedural DMLs

Q5. Differentiate between:
a) Procedural and non-procedural DMLs 2019M

	PROCEDURAL DMLs	Non-Procedural DML
	It is a type of DML that specifies how data is transformed, rather than just what is tobeT. It involves writing code to manipulate data	Its a type of DML which specifies what data is tobe Transformed but not how. It involves specifying declarative statements acc to need.
	It allows for greater control and flexibility as its used for complex operations with loops etc	It is easier to understand and allows for easier maintenance and modification of process.
Eg.)	PL/SQL in Oracle T-SQL in SQL Server	SQL, XQuery, SPARQL

Q5 (a) Write the SQL queries for the relation 2017M 3

Books(title,author,book_id,year_publication)

(i) Books whose author name starts with 'M' but does not end with 'M'.
(ii) Book with the second largest year of publication.
(iii) Books published between 1990-2000 but not in year 1990 and 2000

i.) SELECT * FROM Books WHERE author LIKE 'M%' AND author NOT LIKE '%M'.

ii.) SELECT * FROM Books WHERE

iii.) SELECT * FROM Books WHERE year_publication BETWEEN 1991 AND 1999;

Q5 (a) Write the SQL queries for the relation 2017E 3

Books(title,author,book_id,year_publication)

(i) Books whose author name starts with 'M' but does not end with 'M'.
(ii) Book with the second largest year of publication.
(iii) Books published between 1990-2000 but not in year 1990 and 2000

(b) Consider the following relations : 2016E 4
Customer (Cust_No, Sales_Person_no, City)
Sales_Person(Sales_person_no, Sales_Person_Name, Common_Prec, Year_of_Hire)
Write SQL
(i) Display the list of all customers by Cust_No with the city in which each is located.
(ii) List the names of the sales persons who have accounts in Delhi 6

i) SELECT Customer.Cust_No, Customer.City
FROM Customer
JOIN Sales Person
ON Customer.Sales_Person_no = Sales_Person.Sales_Person_no;

ii) SELECT Sales_Person_Name
FROM Sales_Person JOIN Customer ON Sales_Person.Sales_Person_no = Customer.Sales_Person_no WHERE City = 'Delhi';

(b) Write statements for creating university database including primary and foreign keys, constraints and inserting values in each table. 2016M 3

```
CREATE TABLE STUDENTS (
  student_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name    "            ,
  graduation_year INT,
  CONSTRAINT chk CHECK (gra_yr >= 1900 AND " <= 2100)
);

CREATE TABLE COURSES (
  C_id INT PRIMARY KEY,
  C_name VARCHAR (50) NOT NULL,
  Credits INT
);

CREATE TABLE Enrollments (
  e_id INT PRIMARY KEY,
  Student_id INT NOT NULL,
  C_id   "    ,
  grade VARCHAR (2),
  CONSTRAINT fk_stu FOREIGN KEY (student_id)
    REFERENCES STUDENTS ("),
  CONSTRAINT fk_course " (C_id)
    " COURSES (C_id)
);

INSERT INTO STUDENT VALUES
  (1, "John", "Smith", 2025),
  (2, "Aoi", "Lee", 2024),
  (3, "Jade", "Johnson", 2025),
);

INSERT INTO COURSES VALUES
  (1, "COA", 4), (2, "DBMS", 4), (3, "OS", 4);

INSERT INTO Enrollments VALUES
(1,1,1,B), (2,1,3,A), (3,2,2,C), (4,3,1,A)
;
```

Date / / Page No.

(b) What is nested SQL? What are the advantages of nested SQL programs? Give example. 2016M 5

It refers to the use of one SQL query within another query. It can be accomplished by using subqueries, which are embedded within WHERE and HAVING clause. The inner query is executed first and returns a result set, which is then used by the outer query to produce a final result set.

ADVANTAGES:

→ They are simpler and more consise than writing separate queries especially when querying complex data structures.

→ They are more efficient as they reduce the amound of data transferred between DB and application.

→ They can be used to perform complex operations like filtering, aggregation and sorting.

→ They are easier to maintain as they are more moduler and easy to understand

Eg)

```
SELECT *
FROM customers
WHERE customer_id IN (
        SELECT customer_id
        FROM orders
        WHERE price > 2000;
)
ORDER BY first_name;
```

iv. Write a brief note over SQL. 2019E

Structured Query Language is a standard declarative language for RDBMS. It allow users to communicate with RDBs and insert/update/retrive data from database.

SQL is based on Tuple Relational Calculus which is based on relational algebra.

Commercial systems like Oracle, SQL Server, MS Access etc. provides major features. It was implemented by IBM as SEQUEL in project R in 1970s and became a standard of ANSI in 1986 and ISO in 1987

It consists of:

→ Data Control Language (DCL)
→ Data Defination Language (DCL)
→ Transaction Control Language (TCL)
→ Data Manipulation Language (DML).

b) Define a trigger that inserts a row in a books_duplicate table(with the same schema as Books) whenever a delete is performed over Books relation. 3 2017E

efine a trigger that inserts a row in a books_duplicate table(with e same schema as Books) whenever a delete is performed over ooks relation. 3 2017 M

The trigger can be defined as follows in SQL:

```
CREATE TRIGGER books-delete-trigger
AFTER DELETE ON Books
FOR EACH ROW
INSERT INTO books-duplicate
SELECT * FROM Books WHERE id= OLD.id;
```

Q6 What is the difference between Trigger and Assertion? Give examples. 4 2016E

Trigger is a database object which fires when an event occurs in a database. They are defined in table

Assertion in a statement that ensures certain conditions are met and enforced in the database no matter what. They are defined separately from the table and condition is check on DB which on failure will generate an error

	ASSERTION	TRIGGER
Execution	It always executes when a event occurs at the time of Data manipulation	Executes before/after a specific event occurs.
ACTION	It reports errors and enforces constraints	It can perform pre defined actions on data
SCOPE	Can be defined on tables, views	Can also be defined on other DB objects

```
CREATE TABLE STUDENTS (
    Sid_id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT;
    CHECK (age >= 18)
);
```

```
CREATE TRIGGER update-age BEFORE INSERT OR
UPDATE ON STUDENT FOR EACH ROW
BEGIN
    IF new.age<18 THEN SET new.age =18;
ENDIF;
END;
```

Date / /
Page No.

UNIT 3: DATABASE DESIGN.

[b] What is functional dependency? Explain the inference rules for functional dependency. 2012E 7

Functional dependency is a fundamental concept in database design which is used to describe the relationship between attributes in a table. It is a constraint b/w two sets of attributes in a relation, where the "dependant" attribute is determined by another set of attribute i.e. "determinant".

They can help identify key attributes and can guide the process of normalization.

INFERENCE RULES

The collection of inference rules called Armstrong's Axioms are:

→ REFLEXIVITY RULE: If α is a set of attributes and $\beta \subseteq \alpha$ then $\alpha \to \beta$ holds.

→ AUGMENTATION RULE: If $\alpha \to \beta$ holds and γ is a set of attributes, then $\gamma\alpha \to \gamma\beta$ holds

→ TRANSITIVITY RULE: If $\alpha \to \beta$ holds and $\beta \to \gamma$ holds, then $\alpha \to \gamma$ holds.

→ UNION RULE: If $\alpha \to \beta$ holds and $\alpha \to \gamma$ holds, then $\alpha \to \beta\gamma$ holds.

→ DECOMPOSITION RULE: If $\alpha \to \beta\gamma$ holds, then $\alpha \to \beta$ holds and $\alpha \to \gamma$ holds.

→ PSEUDOTRANSITIVITY RULE: If $\alpha \to \beta$ holds and $\gamma\beta \to \delta$ holds, then $\alpha\gamma \to \delta$ holds.

Q2. (a) Use Armstrong's axioms to prove the soundness of the pseudotransitivity rule. 2019S (5)

if $\alpha \to \beta$ and $\gamma\beta \to \delta$ then $\gamma\alpha \to \delta$

∴ Consider

1.) $\alpha \to \beta$ (Given)

2.) $\alpha\gamma \to \gamma\beta$ (Augumentation Rule)

3.) $\gamma\beta \to \delta$ (Given)

4.) $\alpha\gamma \to \delta$ (TRansitivity Rule on 2-3)

Thus, pseudotransitivity rule is sound.

c). Use Armstrong's axioms to prove the soundness of the ~~pseudotransitivity~~ rule. 2019S (5)

Decomposition

If $\alpha \to \beta\gamma$, then $\alpha \to \beta$ and $\alpha \to \gamma$

∴ consider

1.) $\alpha \to \beta\gamma$ (Given)

2.) $\beta\gamma \to \beta$ (Reflexitivity Rule)

3.) $\alpha \to \beta$ (Transitivity Rule)

4.) $\beta\gamma \to \gamma$ (Reflexitivity Rule)

5.) $\alpha \to \gamma$ (Transitivity)

Using 3,5, Thus, decomposition is sound

Date / /
Page No.

c. Anomalies classification with example 2019E
v. Discuss anomalies and its classification. 2018E

An anomaly refers to an unexpected or unintentional outcome that arises when data is manipulated or processed. They occur when data is inserted, deleted or updated leading to data inconsistency, loss or corruption.

1. INSERTION ANOMALY: It occurs when its not possible to add data to a table without also adding additional data that is not relevant to record
   Eg.) we must include nulls or other values if dept is not reqd in a new employee record.
   It can also occur in a scenario when inserting a new dept. having no employees.

2. DELETION ANOMALY: It occurs when deletion from a table also results in the unintended loss of other data that may be imp.
   Eg) If we delete from emp-dept a tuple then if he was the last employee of that dept, then info of the dept too is lost.

3. UPDATE/MODIFICATION ANOMALY: It occurs when updating data in a table results in inconsistencies or errors.
   Eg.) If manager of a dept change we have to update is records of all employees having that dept.

b) Compute the canonical cover of the following set of functional dependencies for relation schema
R = (A, B, C, D, E). 2019 E (5)

A → BC
CD → E
B → D
E → A

Compute the closure for relational schema 2016M
R={A,B,C,D,E}
A-->BC
CD-->E
B-->D
E-->A
List candidate keys of R.

(b) Consider the following relation schema and set F of functional dependencies
R = (A, B, C, D, E). 2019M (5)
A → BC
CD → E
B → D
E → A
List the candidate keys for R.

(b) Compute the closure of the following set F of functional dependencies for relation schema
R = (A, B, C, D, E). 2019S
A → BC
CD → E
B → D
E → A
List Candidate keys of R (5)

Closure for A

Iteration	Result	Using
1	A	
2	ABC	A→BC
3	ABCD	B→D
4	ABCDE	CD→E

$A^+ = \{A, B, C, D, E\}$

Closure of CD

ITERATION	RESULT	USING
1	CD	
2	CDE	CD→E
3	ACDE	E→A
4	ABCDE	A→BC

$CD^+ = \{ABCDE\}$

Closure of B

IT	RES	USING
1	B	
2	BD	B→D

$B^+ = \{B, D\}$

Closure of E

ITERATION	RESULT	USING
1	E	
2	AE	E→A
3	ABCE	A→BC
4	ABCDE	B→D

$E^+ = \{A, B, C, D, E\}$

$C \to \{C\}$ $D \to \{D\}$

[b] Compute the closure $\{A\}^+$ of relation schema 2012E
R = {A, B, C, D, E}
A → BC, CD → E, B → D, E → A.

∴ Clearly, candidate keys are A, CD, E

(b) Write the algorithm steps to find the closure of a set. Compute the closure of the following set F of functional dependencies for relation schema R = {A, B, C, D, E}. A→ BC, CD → E, B → D, E → A.
2018E (2X7=14)

2018M
Q4 Explain Inference Rules in functional dependency. For a relation R{A,B,C,D,E,F,G} the given set of functional dependencies are as follows A → B, BC → DE, AEF → G. Prove that ACF → DG. (6)

1. A → B (given)
2. BC → DE (given)
3. AEF → G (given)
4. AC → A (reflexivity)
5. AC → AB (augmentation using 1)
6. AC → AE (augmentation using 2)
7. AC → ABDE (using 5 and 2)
8. AC → ABCDEFG (" using 3)
9. ACF → C (reflexivity)
10. ACF → AC (augmentation using 9)
11. ACF → ABDE (transitivity using 10 and 7)
12. ACF → ABDEG (aug using 8)
13. ACF → DG (transitivity using 12 and 7)

?

Date / /
Page No.

Q4 (a) Explain inference rules for functional dependency. For a given relation R{ABCDEF} given that {A→BC, B→E, CD → EF}.Using the rules of functional dependency prove that AD → F. 2018S

1 A→BC (given)
2 B→E (given)
3 CD→EF (given)
4. A→C (decomp of 1)
5. AD→CD (aug of 4)
6. AD→EF (transitivity 5,3)
7. AD→F (decomp of 6

Q5. Write short notes on the following: (any four) (4x2=8)
a) Minimal Cover 2020 M

A minimal cover of a set of functional dependencies E is a minimal set of dependencies (in the standard canonical form and without redundancy) that is equivalent to E. We can always find at least one minimal cover F for any E.
Thus its the smallest set of func. dep. that must be preserved in order to prevent anomalies in DB.

A set of functional dependencies F is minimal if:
→ Every dep in F has single attribute for RHS.
→ we cant replace any dep X→A with Y→A where Y⊆X, in F
→ we cant remove any dep in F.

(b) Given are the two sets of FDs for the relation F and G: 2018M
F = { B → CD, AD → E, B → A} and
G = { B → CDE, B → ABC, AD → E}.
Compute whether they are equivalent sets of FDs or not.

i.) Check if F covers G

for G and taking closure from F

$B^+ = ABCDE$ $\quad$ F = B → CD, AD → E, B → A

$AD^+ = E$ $\quad$ G = B → CDE, B → ABC, AD → E

So F covers G ⇒ $F \supseteq G$ — ①

ii.) Check if G covers F

for F and taking closure from G

$B^+ = ABCDE$

$AD^+ = E$ $\quad$ F = B → CD, AD → E, B → A.

so, G covers F ⇒ $G \supseteq F$ — ②

from ① and ② $F \equiv G$ so they are equivalent.

(d) Write the algorithm steps to determine X^+, the closure of X under F. 2012E

Input: A set F of FDs on relation schema R and a set of attributes X, which is a subset of R.

∴ algorithmic steps.

```
X+ := X;
repeat
old X+ := X+;
for each functional dependency Y → Z in F do
    if X+ ⊇ Y then X+ := X+ ∪ Z;
until (X+ = old X+);
```

Date / /
Page No.

Q4 (a) Define normalization. A company wants to store the complete address of each employee, they create a table named employee_details. Make this table complies with 3NF. 2019E

p_id (A)	emp_name (B)	emp_zip (C)	emp_state (D)	emp_city (E)	emp_dist (F)

Normalization of data can be defined as a process of analyzing the given relation schema based on their FD and primary keys to achieve the desirable properties of
→ minimising redundancy
→ minimising insertion, deletion, updation anomalies

$R(A, B, C, D, E, F)$ clearly CK is A

$\therefore A \rightarrow BC$
$C \rightarrow DEF$

i) check BCNF → fail as C is not CK
ii) check 3NF → $(C \rightarrow DEF)$ violates as C isn't a key and, DEF is non prime
iii) check 2NF → Satisfy the 2NF

∴ converting to 3NF we need to decompose
$\therefore A \rightarrow BC$ and $C \rightarrow DEF$

comply [R1 (ABC)
R2 (DEF)] dont have transitive FD

Date / /
Page No.

Q4. a) Explain the normal forms due to functional dependencies. (5)

Normal forms in RDBs are a set of rules to ensure data integrity and reduce redundancy. The most commonly used normal forms are based on FDs which are:-

1. FIRST NORMAL FORM (1NF):
A table is in 1NF if it does not contain repeating groups and arrays. Each attribute in the table must have an atomic value, meaning it cannot be broken into smaller atoms.
i.e it should not have values separated by commas.

10

2. SECOND NORMAL FORM (2NF):
A table is in 2NF if it is in 1NF and every non-key attribute is fully dependent functionally on the PK. (it must be dependent on the entire PK and not a part of it).

3. 3NF | 4. BCNF (refer 2 page ahead)

Q3 What is the need of normalisation. Discuss 1NF, 2NF and 3NF with example. Consider the following schema and normalize it. (8)

2018E

emp_id	emp_name	emp_zip	emp_state	emp_city	emp_district

Normalisation is needed due to the following reasons:

1. MINIMIZE REDUNDANCY: It allows us to avoid duplicate data and thus conserving space and reduce inconsisting.
2. AVOID ANOMALIES: It helps us to eleminate anomalies during updation or deletion. Eg:) when a record is deleted, the data in other records remain same.
3. IMPROVE INTEGRITY: It is done by enforcing constraints on data. This ensures that data entered into DB is accurate and confirms to predefined rules.
4. FACILITATE DB MAINTENANCE: It simplifies database by organising data in a logical and consistent manner.

[b] Write a code of PL/SQL block to find the largest of three given numbers.

2012E

```
DECLARE
    a number;
    b number;
    c number;
BEGIN
    dbms-output.put-line ('Enter a:');
        a:=&a;
    dbms-output.put-line ('Enter b:');
        b:=&b;
    dbms-output.put_line ('Enter c:');
        c:=&c;
    if (a>b) and (a>c) THEN
        dbms_output.put_line ('A is greatest' ||A);
    elsif (b>a) and (b>c) THEN
        dbms_output.put-line ('B is greater' ||B);
    else
        dbms-output-put-line ('C is greatest' ||C);
    end if
END;
```

[c] How is BCNF different from 3NF? 2012E

(h) Define BCNF and 3NF. Explain the difference with examples. 2016M

b. BCNF vs. 3NF with example 2010E

d) BCNF and 3NF 2019E

Date / / Page No.

3NF:

Third Normal Form is based on the concept of transitive dependency. A relation schema R is in 3NF if it satisfies 2NF and no non prime attributes of R is transitively dependant on the primary key.

An FD is $(X \to Y)$ is a transitive dependency if there is Z, that is neither CK nor a subset of any key R, and both $X \to Z$ and $Z \to Y$ hold.

We can say that, A relation R is in 3NF if whenever a nontrivial functional dependancy $X \to A$ holds in R, either

a.) X is a SK of R

OR, b.) A is a prime attribute of R.

BCNF (Boyce-Codd Normal Form)

A relation schema R is in BCNF if whenever a nontrivial functional dependancy $X \to A$ holds in R, then X is a superkey of R.

Every relation in BCNF is also in 3NF (and not vice versa) ~~The~~

The difference is that in 3NF, A is allowed to be prime but not in BCNF. In practice, most schemas that are in 3NF are also in BCNF. Only if $X \to A$ holds in reln scheme R with X not being a superkey and A being a prime attribute will R be in 3NF but not in BCNF.

Eg.)

7[a] You are given the below functional dependencies for relation R (A,B,C,D,E) 2012 E

F = {AB → C, AB → D, D → A, BC → D, BC → E}

(i) Is this relation is in BCNF? If not, show all dependencies that violate it.

(ii) Is this relation is in 3NF? If not, show all dependencies that violate it. 7

B

Given R(A, B, C, D, E)

AB → C

AB → D

D → A

BC → D

BC → E

to find ck

Clearly.

$AB^+ = ABCDE$

$BC^+ = ABCDE$

$BD^+ = ABCDE$

∴ ck → AB, BC, BD.

i) Check BCNF

No, its not in BCNF

D → A violates - as D is not a candidate key.

ii) Check 3NF

It is in 3NF as all FD satisfy conditions.

Q2 Examine the table below and answer the following: 3x2

staffNo	branchNo	branchAddress	name	position	hoursPerWeek
S4555	B002	City Center Plaza, Seattle, WA 98122	Ellen Layman	Assistant	16
S4555	B004	16 – 14th Avenue, Seattle, WA 98128	Ellen Layman	Assistant	9
S4612	B002	City Center Plaza, Seattle, WA 98122	Dave Sinclair	Assistant	14
S4612	B004	16 – 14th Avenue, Seattle, WA 98128	Dave Sinclair	Assistant	10

a) What is the existing normal form of the table? IT 2020 M

b) Convert this to highest normal form possible.

c) Identify the candidate key/primary key with the help of closure

a) FD

Staff No → name, position

branch No → branch Address

Its in 1NF

ck → {Staff No, branch No}

b)

branch No	branch Address
B002	City ———
B004	16-14th ———

Staff No	name	Pos
S4555	Ellen	Assist
S4612	Dave	"

Staff No	branch No	hours

Q.4 Consider the following relation for published books
Book(Book_title,Author_name,Book_type,List_price,author_affil, publisher)
Author_affil refers to affiliation of author. Suppose the following dependencies exist:-
Book_title -> publisher, book_type
Book_type -> list_price
Author_name -> author_affil
a) What NF is the relation in? Explain your answer. 2017 M 2
b) Apply normalization until you cannot decompose the relation further. State the reason for each decomposition. 2
c) Find the candidate keys using implied rules 2

Book = R
Book_Title = A
Author_Name = B
Book_type = C
List_price = D
author_affil = E
publisher = F

$\therefore R(A, B, C, D, E, F)$

FDs
$A \rightarrow C, F$
$C \rightarrow D$
$B \rightarrow E$

c) find candidate key:
$A \rightarrow CF \mid B \rightarrow E \mid C \rightarrow D$
$\therefore AB \rightarrow ABCEF$
$(AB) \rightarrow ABCDEF$
$\therefore$ AB is CK

a) check BCNF
Its not in BCNF as $C \rightarrow D$, C is not a CK
Check 3NF
Its not in 3NF as in $A \rightarrow CF$ CF are non prime
Check 2NF
partial dependency exist
eg) $B \rightarrow E$ but AB is CK
$\boxed{\therefore \text{Its in 1NF}}$

b) 2NF
to eleminate PD decompose as.
$A \rightarrow CF$
$B \rightarrow E$

R1(A, B)
R2(A, F, C, D)
R3(B, E)

3NF
R1(A, B)
In R2, transitive dep exist $(A \rightarrow C, C \rightarrow D)$
R2a(A, F, C)
R2b(A, D)

Q4. Consider the following relation schema and set of functional dependencies. 2020M (5x2=10)

R = (A,B,C,D,E,F,G,H).

BE → GH
G → FA
D → C
F → B

a) Can there be a key that does not contain D? Explain.
b) Is the schema in BCNF? Explain.
c) Use one cycle of BCNF algorithm to decompose R into two sub-relations. Are the sub-relations in BCNF?
d) Show that decomposition is lossless
e) Is your decomposition dependency preserving? Explain.

a.) No, since D is not uniquely determined by any other attribute, it must be part of all keys??

b.) No, it is not in BCNF since none of the left hand sides are superkey here F → B violates condition as F is not a superkey but still determines B.

c) one cycle of BCNF algo requires finding a non trivial func dep which violates BCNF (Eg) F → B)

$\therefore R_1 = (F, B)$ F → super key
$R_2 = (A, C, D, E, G, H)$

But R_2 isnt in BCNF as its not in 3NF

d.) Decomposition is lossless because there are not any FD b/w the attributes in (F,B) and (A,C,D,E,G,H). (so no addn tuples)

Natural Join $[(F,B) \bowtie (A,C,D,E,GH) = R]$ will produce all tuples in R

e) It is dependancy preserving because each of the FDs of R is a FD of one of component relations

(b) Consider the Universal relation R(A,B,C,D,E,F,G,H,I,J) and set of FD's G=({A,B}->{C}, {B,D}->{E,F}, {A,D} -> {G,H}, {A}->{I}, {H} -> {J}) What is the key of R? Decompose R into 2NF. then 3NF relations. 8 17E

R (A, B, C, D, E, F, G, H, I, J)

FD

FD	
AB → C	to find CK
BD → EF	A^+ = AI ✗
AD → GH	AB^+ = ABCEFJ
A → I	ABD^+ = ABCDEFGHIJ ✓
H → J	ABD is candidate key of R

It is in 1NF

1.) decompose to 2NF

we will eleminate PD

AB → C | BD → EF | AD → GH, H → J | A → I

∴ R1 (ABC)

R2 (BDEF)

R3 (ADGHJ)

R4 (AI)

R5 (ABD)

2.) Decompose to 3NF

we will eleminate transitive dependency

R1 (ABC)

R2 (BDEF)

R3a (ADGH)

R3b (HJ)

R4 (AI)

R5 (ABD)

b) Disjoint and overlapping generalization

2019M

b) Disjoint and overlapping generalization

Date / /
Page No.

Q2.Using a two-dimensional array of size n x m, illustrate the differences (a) in the three levels of data abstraction and (b) between a schema and instances. (4)

2019M

Date / /
Page No.

vii. Discuss join dependencies.

2019(E) old

A join dependency (JD), denoted by JD($R_1, R_2, \ldots R_n$) specified on relation schema R, specifies a constraint on the states r of R. The constraint states that every legal state r of R should have a non additive join decomposition into $R_1, R_2 \ldots R_n$. Hence for every such r, we have

$$*(\pi_{R_1}(r), \pi_{R_2}(r), \ldots \pi_{R_n}(r)) = r$$

Thus, a join dependency specifies that a set of tables must always be joined together in queries and that they cannot be separated. This constraint ensures that the data in tables is not duplicated or lost when performing queries that require multiple tables to be joined.

It is important to eleminate JD to achieve 5NF.

(b) Define the following: 2017E 8
(i) Fully function dependency
(ii) Multivalued dependency
(iii) Trivial dependency. 5

i.) A fully functional dependency is a relationship between two or more attributes in a table where knowing the value of one attribute uniquely determines the value of another attribute.
∴ If A determines B and there are no other attributes that can determine B, then A has a fully functional dependency on B.

ii.) A MVD is a type of dependency between two sets of attributes in a table where for each unique value of one set of attributes, there can be multiple values of for the other set of attributes.
Eg) one writer can author mutible books.

iii) A trivial dependency is a dependency where the determining attribute is already a part of the determined values. i.e. the dependency is always true w/o any additional info. Eg) $AB \rightarrow A$.

UNIT 4.2: Transaction Processing Concepts
UNIT 5: Concurrency Control Techniques.

Date / /
Page No.

Q1 Attempt the following:
vi. Explain ACID properties. 2019E

i. Discuss ACID properties in transactions with example. 2018E

Q3. a) List the ACID properties. 2019S

4[a] Explain ACID properties of transaction. Also draw the state transition diagram illustrating the states for transaction execution. 2012E 7

ACID properties are some properties that transactions possess which are enforced by the concurrency and control and recovery methods of the DBMS. They are

→ ATOMICITY: A transaction is an atomic unit of processing; it should either be performed in its entirety or not performed at all.
Eg.) In a bank system, if money is to transfered, then it should be either debited from one acc. and credited to another as a whole or not at all.

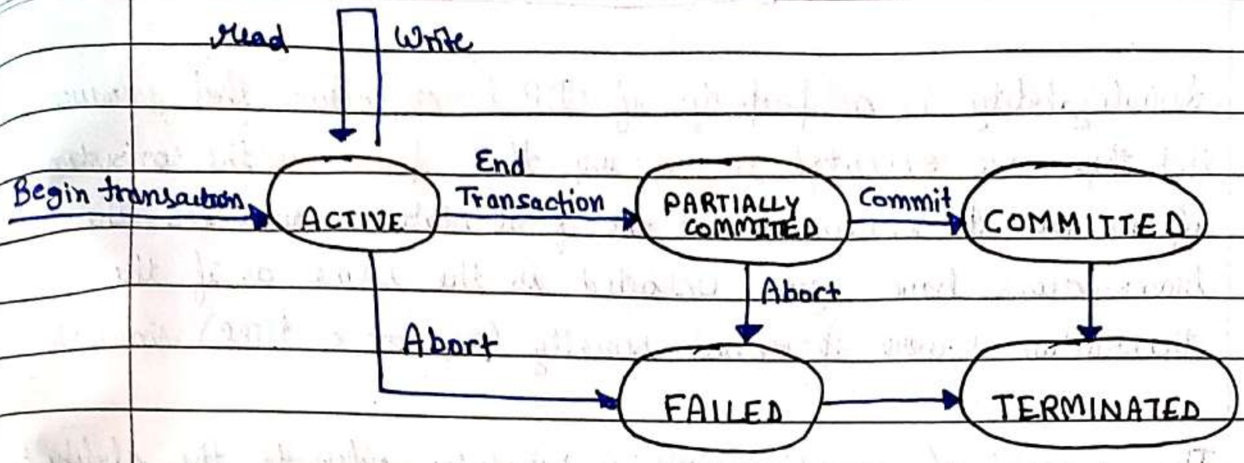
→ CONSISTENCY: A transaction should be consistency preserving, i.e. if it is completely executed from beginning to end w/o interference from other instructions, it should take the DB from one consistent state to another.
Eg.) If a transaction attempts to debit an empty account, it should be rolled back and returned to prev state.

→ ISOLATION: A transaction should appear as though it is being executed in isolation from other transactions, even though many transactions are being executed concurrently. i.e., it should not be interfered with by any other transaction.
Eg.) If two transactions attempt to change the same record in DB, then isolation property ensures no conflict.

→ DURABILITY: The changes applied to the database by a committed transaction must persist in the DB. These changes must not be lost because of any failure.
Eg.) It ensures any updates in records of DB will remain after restart

Date / / Page No.

vi. Draw the life cycle of a transaction with all possible states. 2018E



State Transition Diagram showing various states for transition execution

c) What is cascade less schedule? Why cascadelessness of schedules desirable? Explain your answer. 2019S (5)

In a cascadeless schedule, if any of the transactions have performed the write operation on an item then other transactions are not allowed to read that data item untill the transaction that updated the data item is not committed or aborted.

Ex)

T_1	T_2	T_3
R(A)		
R(B)		
W(A)		
W(B)		
Commit		
	W(A)	
	W(B)	
	Commit	
		R(B)
		W(B)
		Commit

The cascadeless schedules are desirable because if a schedule is not cascadeless and one transaction failed then all of the dependent transactions will be aborted or ~~rolled~~ roll back.

Thus here, failure of a t/r does not lead to aborting of any other t/r.

Date / /
Page No.

[b] Explain the concept of serializability of schedules. 2012E — Serializability 2017E

Serializability is a property of DB transactions that ensures that they are executed in a way that preserves the consistency of the DB. It ensures that the final state of DB after the transactions have been executed is the same as if the transactions were processed serially (one at a time) in order.

The concept of serializability of schedules refers to the ability to ensure that concurrent trs appear to be processed serially.

NOTE: If no interleaving is allowed then if 2 trs are present.

so,

S_1

T_1	T_2
	R(B) W(B)
	R(A) W(A)
R(A) W(A)	
R(B) W(B)	

Serial

S_2

T_1	T_2
R(A) W(A)	
	R(B) W(B)
R(B) W(B)	
	R(A) W(A)

Non Serial but serializable

(b) Explain serializability of schedules. What is a conflict and view serializable schedule. 2019E

∴ A Schedule is serializable if it is equivalent to some serial schedule of the same transactions.

(c) When the two schedules are conflict equivalent? Explain. 2017E 2

(1) CONFLICT SERIALIZABLE

A schedule is said to be conflict serializable if it is conflict equivalent to a serial schedule. Two schedules are said to be conflict equivalent if the order of any two conflicting instructions is the same in both schedules.

Thus, in a conflict serializable schedule, order of conflicting operations (i.e. operations that access the same data items where atleast one of them is a write operation) is the same in all possible equivalent serial schedule.

Eg:)

S_1

T_1	T_2
R(A)	
W(A)	
	R(A)
	W(A)
R(B)	
W(B)	
	R(B)
	W(B)

eq to →

S_2

T_1	T_2
R(A)	
W(A)	
R(B)	
W(B)	
	R(A)
	W(A)
	R(B)
	W(B)

② VIEW SERIALIZABLE

A schedule is said to be view serializable if it is view equivalent to a serial schedule.

Two schedules S and S' are said to be view equivalent if the following conditions hold:

i:) The same set of transactions participate in S and S', and they include the same operations of those transactions.

ii:) For any operation $r_i(X)$ of T_i in S, if the value of X read by the operation has been written by an operation $w_j(X)$ of T_j (or if it is the original value of X ~~because~~ before the schedule started), the same condition must hold for the value of X read by operation $r_i(X)$ of T_i in S'.

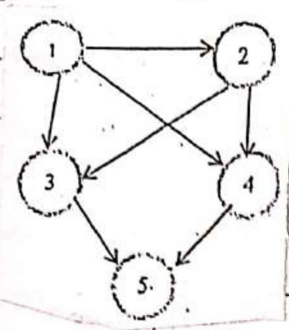
iii:) If the operation $W_k(Y)$ of T_k is the last operation to write item Y in S, then $W_k(Y)$ of T_k must also ~~hold~~ be the last operation to write ~~option~~ item Y in S'.

Thus, a view serializable schedule ensures that the read-write dependencies between transactions are preserved in all possible equivalent serial schedules.

Eg:)

Date / /
Page No.

Q3. a) Consider the following precedence graph. Is the corresponding schedule conflict serializable? Explain your answer. 2019E (3)



We can observe that the following graph is acyclic. If the precedence graph then we can conclude that the schedule is conflict serializable.

∴ order :- T_1 T_2 T_3 T_4 T_5

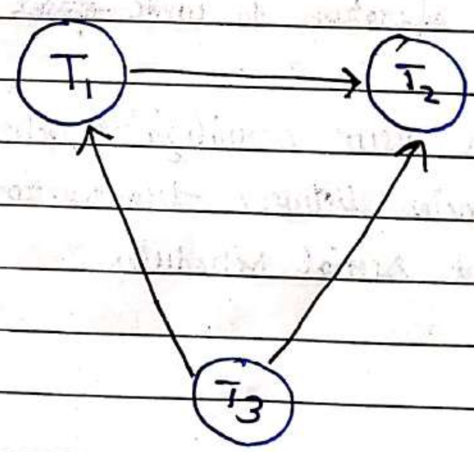
(b) List the algorithm steps for testing serializability of a schedule. Test whether the following schedule is serializable or not :
S_f: $r_3(Y)$; $r_3(Z)$; $r_1(X)$; $w_1(X)$; $r_2(Z)$; $r_1(Y)$; $w_1(Y)$; $r_2(Y)$; $w_2(Y)$; $r_2(X)$; $w_2(X)$;

Steps 2018E (2X4=8)

1. Make a directed graph (create vertices of the transaction).
2. on taking an operation of a transaction look for write operation on another transaction after it |or commit of same variable name.
3. If yes then create a directed edge from $T_1 \rightarrow T_2$
4. Continue till we either exhaust all operations or a cycle is formed.
5. If cycle exists → NOT conflict serializable

S_f

T_1	T_2	T_3
		read (Y)
		read (Z)
read (X)		
write (X)		
	read (Z)	
read (Y)		
write (Y)		
	read (Y)	
	write (Y)	
	read (X)	
	write (X)	



Date / /
Page No.

vii. What are the conditions for conflicting operations in a schedule.

2018E(old)

The three conditions for conflicting operations in a schedule are:

1. The operations are performed by different transactions.

2. The operations access the same data item.

3. At least one of the operations

b) What is recoverable schedule? Why is the recoverability desirable? Explain with suitable example. 2019E (3)

A recoverable schedule is one in which if a transaction T_j reads a data item previously written by a transition T_i, then the commit operation of T_i appears before the commit operation of T_j.

T_8	T_9
read (A)	
write (A)	
	read (A)
read (B)	

It is not recoverable if T_9 commits immediately after read. Thus it should commit after T_8 does so it can too rollback after failure.

Recoverable schedules are desirable because failure of a transaction might otherwise bring the system into an irreversably inconsistent state.

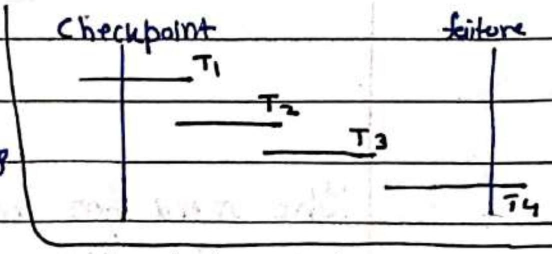
Date / /

Page No.

Q3.(a) Explain various recovery techniques during transaction in detail. 2017E 9

Date: / /
Page No.

Q5. a) Explain the use of Checkpoints in recovery scheme. 2019S(2)



Checkpoints are used in recovery schemes, by reading the transaction log file in reverse order and creating redo and undo lists to recover committed and uncommitted transactions respectively. Transactions with both start and commit operations are stored in the redo list while t/r with only start are in undo list.

o/p	list
<Tn, Start> <Tn, commit>	Redo
<Tn, commit>	Redo
<Tn, Start>	Undo

Ex) in the image

T_1	T_2	T_3	T_4
S			
	S		
	C	S	
		C	
			S Fail

∴ Redo list – T_2, T_3, T_4

Undo list – T_1

c. Types of failure in recovery 2018E

Various types of error that can occur during recovery process:-

1. SYSTEM FAILURE: It occurs when the running system crashes or shuts down unexpectedly. This can cause loss of data not written on disk yet.

2. MEDIA FAILURE (disk failure): It occurs when the storage media containing the DBMS is damaged or corrupted resulting in loss of data in the media.

3. TRANSACTION FAILURE: It occurs when a t/r can't be completed due to a software or hardware failure. Eg) network failure, power outage.

4. USER ERROR: It occurs when a user accidently deletes or modifies data in DB resulting in data loss or corruption.

Date / /
Page No.

Q5 (a) What are the problems that can occur when concurrent execution of transaction takes place. Discuss with example. 2018E

Q4 (a) What is the need of concurrency control techniques. Discuss three major concurrency problems. 2018E

The need for concurrency control techniques is to avoid data inconsistencies that may arise when multiple transactions try to modify the same data simultaneously.

Without concurrency control, the following problems may occur:

→ **LOST UPDATE**: This problem occurs when two transactions that access the same DB items have their operations interleaved in a way that makes the value of some DB item incorrect. So, when both update it simultaneously, only one update is applied. The other updates are lost and final state isnt as desired.

T_1	T_2
R(A)	
W(A)	lost
	W(A)
	C
C	

[1] Explain dirty read problem with example. 2012E

→ **DIRTY READ**: This problem occurs when a transaction reads uncommited data that has been modified by another transaction but not yet commited. If the transaction that modified the data rolls back, then the read data may be incorrect leading to inconsistencies.

T_1	T_2
R(A)	
W(A)	
	R(A) dirty
	C
ROll	
C	

→ **UNREPEATABLE READ**: It occurs when a transaction T reads the same item twice and the item is changed by another transaction T' b/w the two reads. Hence, T recieves different values of its two reads of the same item.

T_1	T_2
R(X)	
	R(X)
W(X)	changed
	R(X)

→ **PHANTOM READ**: It occurs when a transaction re-executes a query returning a set of rows that satisfy a search condn and finds the set of rows satisfying the condn has changed as a result of another recently committed transaction.

Date / /
Page No.

→ INCORRECT SUMMARY: If one transaction is calculating the ~~aa~~ aggregate summary function on a no of DB items while other transactions are updating some of those items, the aggregate function may calculate some values before they are updated and some after they are updated.

(a) What is concurrent control? How is it implemented in DBMS? Explain.
2017E

Concurrency control refers to the techniques used to ensure that multiple transactions executing simultaneously do not interfere with each other and that the DB remains in a consistent state. It is essential in multi user environments, where many users may simultaneously access and update the same database.

It is implemented in DBMS using various techniques such as locking, timestamp ordering, graph/tree based protocol, phase locking, multiversion concurrency control etc.

Q5 (a) Describe different locking techniques of concurrency control. Discuss with example.
2019E

In lock based protocols, we use the following types of locks.

1. Binary locks: A binary lock can have two states or values: locked and unlocked (or 1 and 0): A distinct lock is associated with each database item X.
   Two operations: lock-item and unlock-item are used.

c. Shared/Exclusive locks

2. Shared/Exclusive locks:
   A shared lock allows multiple processes to access a resource simultaneously in a read-only mode. It is commonly used when multiple users need to read the same data, but none of them can modify it.

   An exclusive lock allows only one user to access a resource at a time in a write mode. It is used when a process or user needs to

Date / /
Page No.

modify a resource, to ensure that no other process or user can access the resource at the same time and cause conflicts.

Q5 (a) Explain 2 phase locking for guaranteeing serializability. Discuss its variations. 2018E

Q4. a) Prove that the two-phase locking protocol ensures conflict serializability. 2019E (3)

2PL is a concurrency control protocol that guarantees serializab-ility, it works by dividing the transaction into two phases:

(Expanding)
1.) LOCKING PHASE: Here, the transaction acquire all locks it needs before making any modifications to the database. Once a lock is granted, it cannot be released until the unlocking phase.

(Shrinking)
2.) UNLOCKING PHASE: Here, the transaction releases all the locks it acquired in the locking phase. i.e. it can commit or rollback only after it has released all locks.

It can be proved that, if every transaction in a schedule is guaranteed to be seria→

T_1	T_2
read_lock(Y)	read_lock(X)
read_item(Y)	read_item(X)
write_lock(X)	write_lock(Y)
unlock(Y)	unlock(X)
read_item(X)	read_item(Y)
X = X+Y	Y = X+Y
write_item(X)	write_item(Y)
unlock(X)	unlock(Y)

PROOF

Date / /
Page No.

VARIATIONS

1. Conservative (Static) 2PL: Requires a transaction to lock all the items it accesses before the transaction begins execution by predeclaring its read-set and write-set. It is deadlock-free but difficult in practice due to predeclaration.

2. Strict 2PL: A transaction holds all its exclusive (write) locks until after it commits or aborts, ensuring a strict schedule for recoverability. It is not deadlock-free.

3. Rigorous 2PL: A transaction holds all its locks (exclusive or shared) until after it commits or aborts, making it easier to implement than strict 2PL. It also guarantees a strict schedule.

b) What are advantages of rigorous two-phase locking protocol over other variants of two-phase locking? 2019S (3)

1. Guarantees Strict Schedule: Rigorous 2PL ensures serializability and recoverability by guaranteeing a strict schedule, which means that the transaction order is equivalent to the serial order.

2. Easy to Implement: It is easier to implement than strict 2PL because it allows more concurrency and does not require the transaction to hold exclusive locks on all items until it commits or aborts.

3. No Deadlocks: It does not suffer from deadlocks, unlike strict 2PL, as it allows transactions to release shared locks because acquiring exclusive locks, breaking potential deadlock cycles.

4. Efficient Use of locks: It can make efficient use of locks by allowing multiple transactions to hold shared locks on the same item simultaneously, improving the system throughput and response time.

Date / /
Page No.

(b) Which locking protocols ensure freedom from deadloack and cascadless rollback and how? explain. 2017E 5

c) What is time stamp? Discuss the working of wait -die and wound-wait transaction. 2019S (5)

Time stamp is a piece of information that stores the date and time when an event occured. Each transaction is assigned a timestamp for isolation. For each dataitem, the system keeps two timestamps. The read timestamp holds the largest (i.e most recent) timestamp of those t/n that read that data item. The write timestamp of a data item holds the timestamp of the t/r that wrote the current value of the data item.

Date / /

Page No.

Q4. (a) Explain the time stamp ordering protocol. Is it better than 2P locking protocol? 2017E 8

The timestamp ordering protocol is a method used to provide ordering and consistency in a distributed database system. In this protocol each transaction is assigned a unique timestamp, which reflects the time when it started. The protocol operates as follows:

1. Suppose that transaction Ti issues read (Q).
   a) If TS (Ti) < W-timestamp (Q), then Ti needs to read a value of Q that was already overwritten. Hence, the read operation is rejected, and Ti is rolled back.
   b) If TS (Ti) ≥ W-timestamp (Q), then the read operation is executed, and R-timestamp (Q) is set to the maximum of R-timestamp (Q) and TS (Ti).

2. Suppose that transaction Ti issues write(Q)
   a) If TS (Ti) < R-timestamp (Q), then the value of Q that Ti is producing was needed previously, and the system assumed that that value would never be produced. Hence, the system rejects the write operation and rolls Ti back.

   b) If TS (Ti) < W-timestamp (Q), then Ti is attempting to write an obsolete value of Q. Hence, the system rejects this write operation and rolls Ti back.

   c) Otherwise, the system executes the write operation and sets W-timestamp (Q) to TS (Ti).

Date / /
Page No.

5[a] Discuss the time stamp ordering protocol for concurrency control. How does strict timestamp ordering differ from basic timestamp ordering? 2012E 7

(b) Explain concurrency control based on time stamp ordering. How is it different from multiversion techniques based on time stamp ordering. 2018E (2X7=14)

The timestamp ordering protocol is used to order the transactions based on their timestamps. The order of transaction is thus the ascending order of t/r creation.

It works as follows: after assigning a unique TS to each t/r:

1.) Check the following condition when a t/r T_i issues a Read (X) operation:
- → if $WTS(X) > TS(T_i)$ then operation is rejected.
- → if $WTS(X) <= TS(T_i)$ then operation is executed
- → timestamps of all data items are updated.

2.) Check the following condition when a t/r T_i issues a Write(X) operation:
- → if $TS(T_i) < RTS(X)$ then operation is rejected
- → if $TS(T_i) < WTS(X)$ then operation is rejected and T_i is rolled back otherwise operation is executed.

It ensures serializable order and high concurrency. It may cause waiting queue

In basic TS ordering, transactions are allowed to access data items in any order as long as their time stamps do not conflict. In contrast, strict TS ordering enforces a strict order among transactions based on their TS, even if it leads to unnecessary waiting queue. Thus, Strict TS ordering provides higher level of serialization.

Timestamp ordering protocols use time stamps to determine the order in which transaction can access data items, while multiversion techniques create multiple versions of data items to allow multiple transactions to access them concurrently. Time Stamp ordering protocols provide high concurrency and serializable execution, while multiversion techniques provide more flexible concurrency and allow t/r to access diff. versions of data items.

Date / /
Page No.

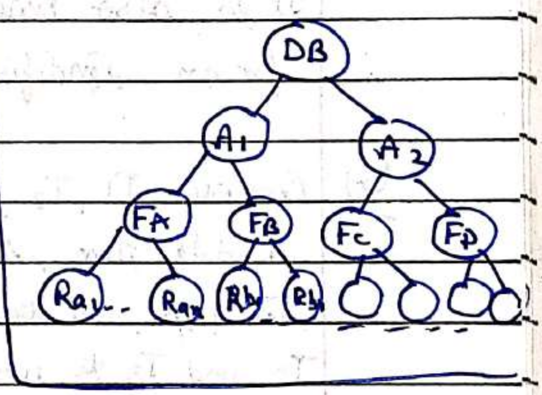
Q4. a) When a transaction is rolled back in timestamp ordering, it is assigned a new timestamp. Explain the reason. 2019 S (2)

A transaction is rolled back because a newer transaction has read or written the data which it was supposed to write. If the rolled back transaction is re-scheduled with the same timestamp, the same reason for rollback is still valid, and the transaction will have be rolled back again. This will continue indefinately.

(e) Multiple Granularity 2017E

(Granularity is the size of data that is allowed to be Locked)
∴ Multiple granularity is the heirarichal breaking up of database into smaller blocks that can be locked.
This locking technique allows locking of various data sizes and sets.
This decreases lock overhead and increases concurrency of DB.

Ex)
1. first level represents entire DB containing several tables and records.
2. second level has nodes that represents areas.
3. Third level contain files present in one area.
4. The bottommost layer in our tree is records which are individual tuples.



There are some limitations to two phase locking that are overcome using Intention mode locks.

1. Intention Shared (IS) lock: node has IS lock only when lower nodes have Shared locks
2. Intention Exclusive (IX) lock: Explicit lock on lower lvl tree with node with shared or exclusive lock.
3. Shared and Intention Exclusive (SIX) lock: subtree is explicity locked in shared mode and with an exclusive mode lock at the lower lvl.

compatibility matrix

	IS	IX	S	SIX	X
IS	✓	✓	✓	✓	✗
IX	✓	✓	✗	✗	✗
S	✓	✗	✓	✗	✗
SIX	✓	✗	✗	✗	✗
X	✗	✗	✗	✗	✗

iv. How a deadlock is different from a live lock. Explain with example. 2018E (4X2=8)

(b) Explain the concept of deadlock. How is it different from starvation. Discuss with the help of an example. 2019E (2X7=14)

A deadlock is a condition where two or more transaction are waiting indefinately for one another to give up locks. It makes a situation where no no task ever gets finished and is in waiting state forever.

Eg) t/r T_1 holds a lock on some rows and needs to update some rows in grade table and simultaneously T_2 needs to update rows in student table held by T1 and is itself holding grade table.

[g] Explain the concept of starvation. 2012E

on the other hand, starvation occurs when a ~~process~~ tr/n is unable to get the reqd. resources even if they are available. It happens when a low priority t/r keeps waiting for resources that are held by a higher priority transaction.

It is also known as a livelock as it has to keep waiting for an indefinate period instead of never like deadlock.

Eg) Consider T1, T2, T3, in a scenario, T_1 is granted a lock on I data item, while T_2 and T_3 are waiting. When T_1 completes, T_4 requests a lock on I and is granted the lock, causing T_2 and T_3 to wait indefinitely leading to starvation.

b) What are the problems caused by deadlocks? 2019S

) What are the problems caused by deadlocks?

It can cause the following problems:

1.) TRANSACTION FAILURE: It can lead to failure causing data inconsistency and make it difficult to recover the DB.

2.) REDUCED THROUGHPUT

3.) INCREASED RESPONSE TIME

4.) INCREASED RESOURCE UTILIZATION

5.) REDUCED AVAILABILITY

Date / /
Page No.

(b) Explain Log based recovery scheme deferred database modification approach

(4) 2017E

Log-based recovery is a technique used in DBMS to ensure data consistency and recoverability in case of failures or crashes. It involves keeping a log of all tr executed on the DB, which includes both before and after state of the DB.

Deferred database modification is an approach where the actual modifications to the database are deferred untill the transaction that made changes is committed. This approach ensures that if a transaction fails or aborts, the changes made by that transaction can be rolled back, and the DB can be restored to its previous consistent state.

In this method, all logs are created and stored in the stable storage, and the database is updated when a transaction commits. The log files are written to disk before the actual modifications is made to database.

UNIT 4.1: File Organization

b. File organization techniques 2019E

(d) File organization 2017E

File organization is a logical relationship among various records. This method defines how file records are mapped onto disk blocks. It is used to describe the way in which the records are stored in terms of blocks, and the blocks are placed on storage medium.

Types of file organization are:
→ Sequential FO
→ Heap FO
→ Hash FO
→ B+ Tree FO
→ Clustered FO.

(b) Explain the concept of indexing. Discuss its classification with diagram and example. 2018E (2X4=8)

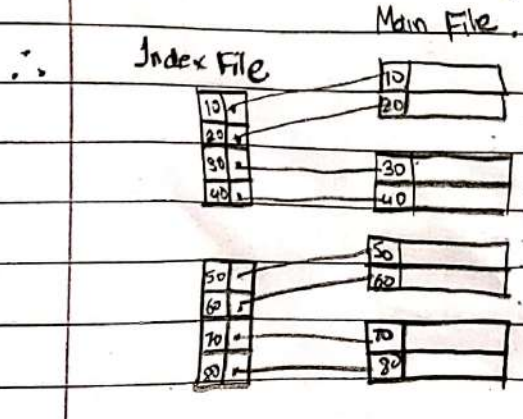
[E] Indexing 2012E

Indexing is a data structure technique to efficiently retrieve records from the database files based on the attributes on which the indexing has been done.
An index file consists of records (called index entries) in the following form:

Search Key	Pointer

Search key : Attribute used to look up records in a file

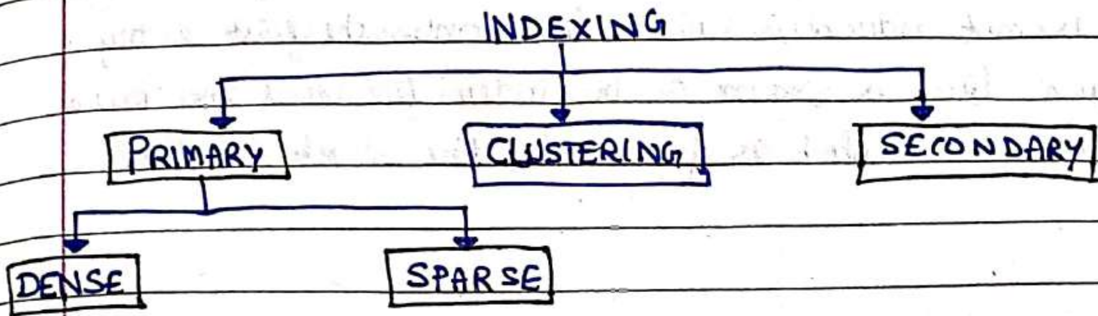
Pointer : Locates the memory address of that attribute.



Date / /
Page No.

(a) Explain different index schemes used in the database system. 2017E

Classification of Indexing



1:) PRIMARY INDEXING

If the index is created on the basis of the primary key of the table, then it is known as primary indexing. As primary keys ~~of~~ are stored in sorted order, the performance of the searching operation is quite efficient. Thus, it is used to enforce the primary key constraint.

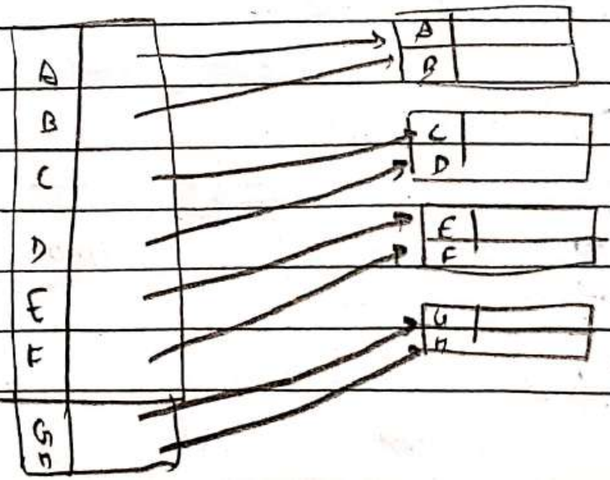
It can be classified into:

c) Sparse and dense Indexing 2019E

i:) DENSE INDEXING:

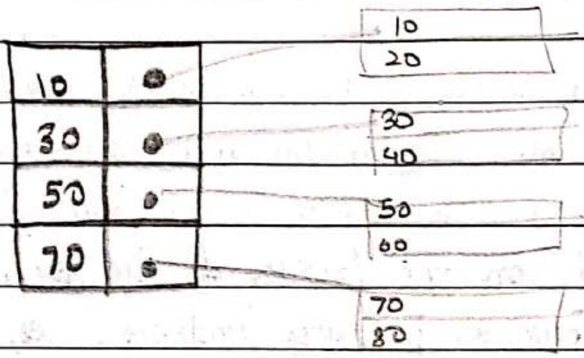
The dense index contains an index ~~for~~ record for every key value in the data file. It makes searching faster. In this, the # records in the index table is same as the # records in the main table.

It needs more space to store the index record itself. The index records have the search key and the pointer to the actual record on the disk.



1 ii:) SPARSE INDEXING

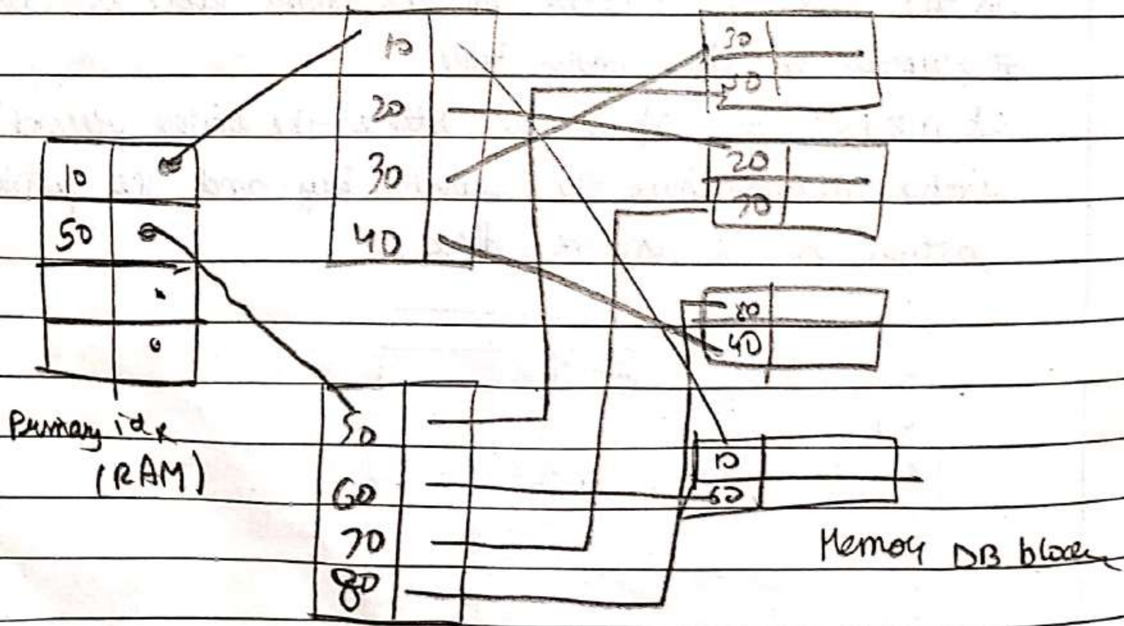
In sparse indexing, each and every value of primary key is not entered in index file. Only the first entry of each block is stored in the index file and the base address of that block is pointed by the block pointer.



2.) SECONDARY INDEXING

In secondary indexing, main file is unsorted. It can be done on key attribute as well as non key attribute. It is used to speed up the retrieval of data based on criteria other than the primary key. It is created on field that are frequently used in queries.

It is not possible on secondary indexing. Because we will have to enter each value in the index table (so dense).



Secondary idx (HD)

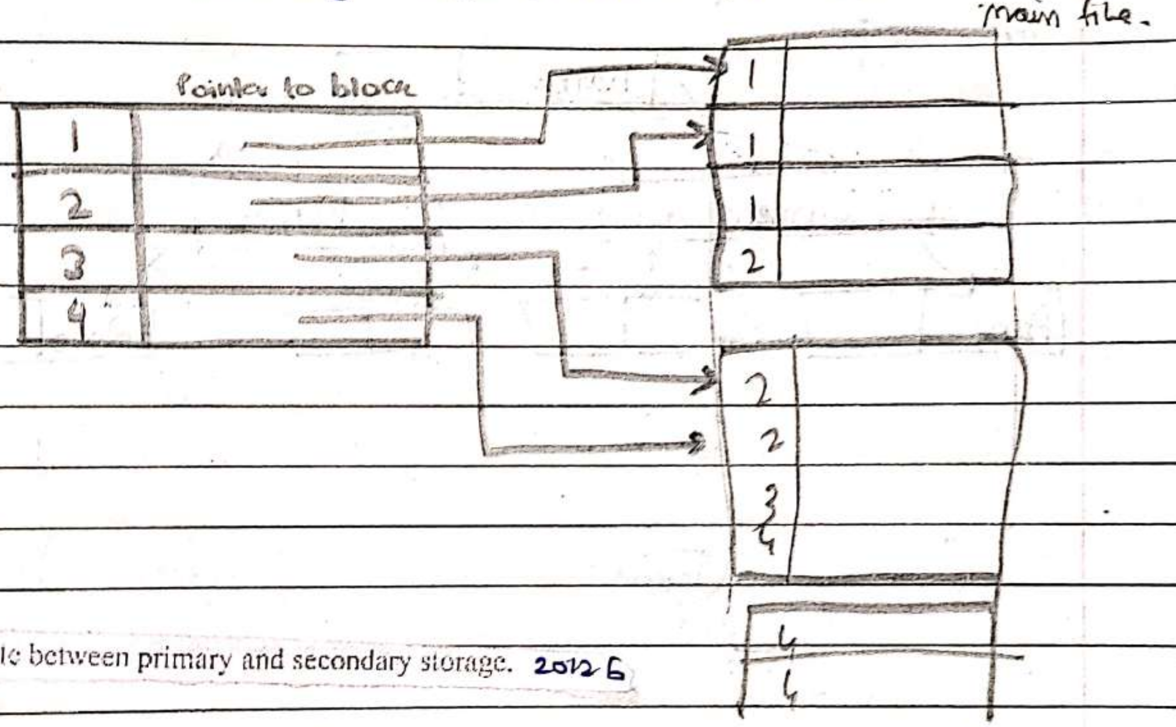
Dense

Date / /
Page No.

3.) CLUSTER INDEXING

Here, main file is sorted using non-key attribute. There will be one entry for each unique value of the non key attribute. Block pointer will point on the first block in which the unique value appears.

Cluster index is an example of sparse index as well as dense index because it doesn't allow every entry in the index file, but it allows every unique value in the index file.



Differentiate between primary and secondary storage. 2012 6

Primary storage refers to the volatile memory used by system to temporarily store data and instructions while the system is running. This includes RAM and cache. Queries are processed here.

Secondary storage refers to the non-volatile memory devices used for long term data storage, such as HDD, SSD. This is where actual DB files are stored and persist after shut down

Date / /
Page No.

Q6 Write short notes (Any two):

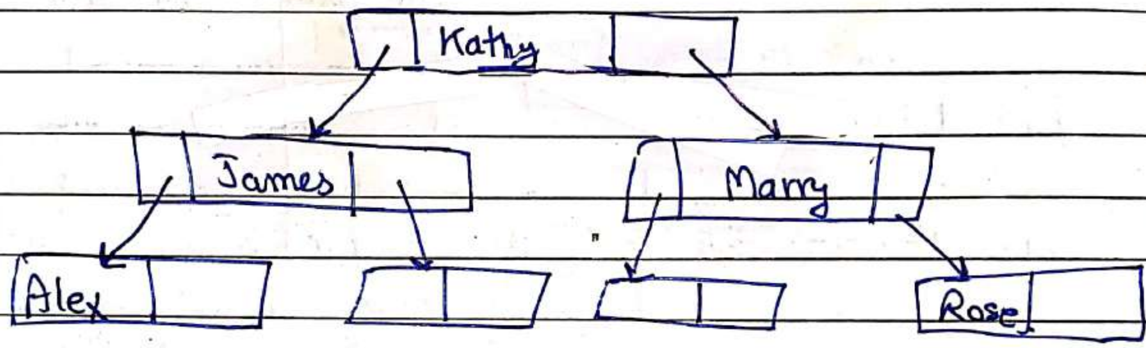
a. B-tree index files = B-Index file 2017 E

B-tree index is a widely used data structure for Indexing. It is a multilevel index format technique which is a balanced binary search tree. All leaves of a B tree signify actual data pointers.

In a B-tree, at every level we are going to have key and data pointer.

Since all intermediary node also have records, it reduces the traversing till leaves for data.

It can be represented as.



insert

Date: / / Page No.

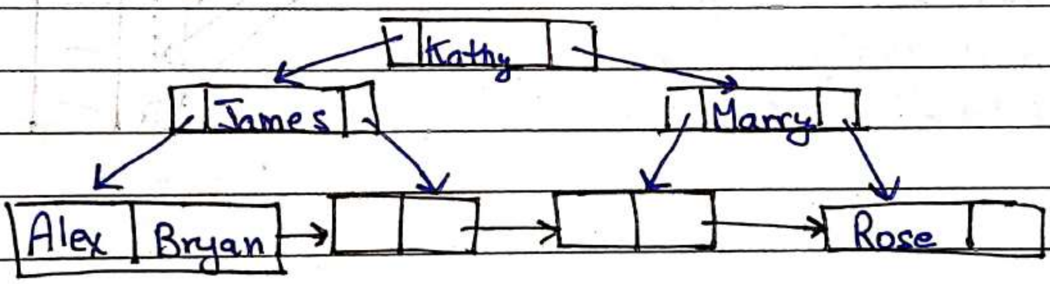
b) Discuss with example how updates are made in B^+ trees. (3) 2019E

A B+ tree is a balanced BST that follows a multi-level index format. The leaves node of a B+ tree denote actual data pointers. B+ tree ensures that all leaf nodes remain at the same height,

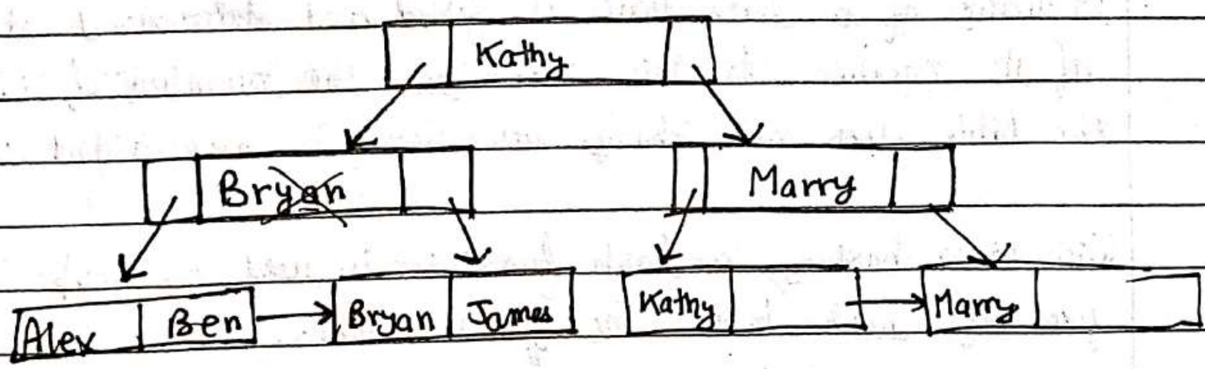
Additionally, the leaf nodes are linked using a linked list; therefore a B+ tree can support random access as well as sequential access.

Insertion/Updation in B+ Tree does not take much time. Hence, B+ tree forms an efficient method to store the records. Incase it finds that tree will be unbalanced because of insert/update/delete, it does the proper re-arrangement of nodes so that the defination of B+ tree is not changed.

Ex:)



If we want to insert Ben, rearrangement is made.

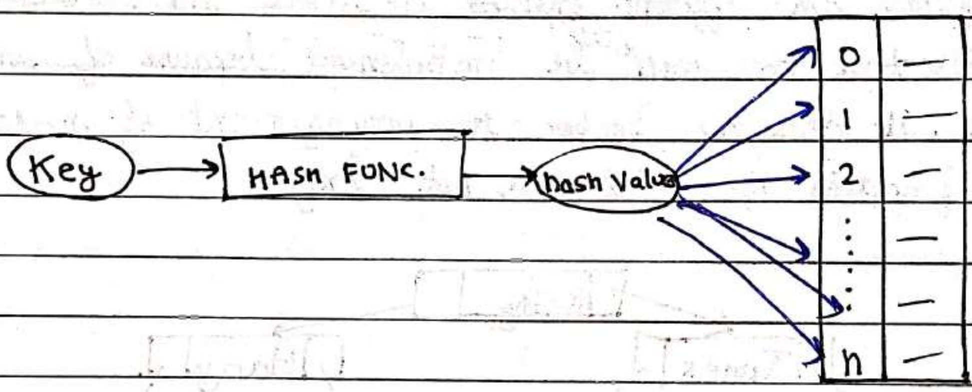


Date / /
Page No.

6. Write short notes (*any two*): 2012E

[a] Hashing Techniques

Hashing is a technique to directly search the location of desired data on the disk without using index structure. Hashing method is used to index and retrieve items in a database as it is faster to search the specific item using the shorter hashed key instead of using its original value. Data is stored in the form of data blocks whose address is generated by applying a hash function in the memory location where these records are stored known as data block or data bucket.



iii. Differentiate between static hashing and dynamic hashing. 2018E

Static hashing is a technique used in DBMS where the size and structure of a hash table is fixed and determined at the time of its creation. In this technique, the number of slots in the table does not change as elements are added or removed.

With static hashing, a hash function is used to map keys to the fixed size hash table. The function takes.

Date / / Page No.

ii. Explain the concept of buffer management. 2018 E(old) [b] Buffer Management 2012 E

A database buffer is a section in the main memory that is used for the temporary storage of data blocks while moving from one location to other. A copy of the disk blocks is kept on the database buffer.

A buffer manager in a DBMS allocates space in main memory for temp data. It sends block addresses to users when data is present in the buffer, and allocates new blocks if they are not found. If the buffer is full, it removes old blocks and writes updates to disk if necessary. If removed data is requested, the buffer manager retrieves it from disk and returns the address. The buffer manager operates independently and its actions are invisible to other programs accessing the disk or buffer.

Buffer management in DBMS applies methods to provide the best services in terms of database buffer management in the main memory which are Buffer Replacement Strategy, Pinned Block and Forced Output of Blocks.

What is a heap file and how the pages are arranged in a heap file? 2017E

A heap file is a type of file organization used for storing unordered records. It is known as heap-organized table. It consists of a set of data pages, each of which contains a fixed number of records. Records are stored in no particular order, and new records are appended to the eof. (there is no index or sorting structure)

The pages in a heap file are arranged in an unordered fashion. The pages themselves are not connected in any particular order. Instead, each page is identified by a unique page no., which is used to locate and retrieve the page from disc when needed.

For new record, if there is enough space on the current page ~~to~~, it is added on that page, otherwise a new page is created on which that record is written.